

Spatial Sampling-based Passivity and Synchronization of Multiweighted Coupled Reaction-diffusion Neural Networks

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Abstract. A type of multiweighted coupled reaction-diffusion neural networks with general delays is formulated in this study, and the passivity and synchronization are addressed via spatial sampling control. Firstly, an innovative spatial sampling controller is developed, which is distributed at the midpoint of each spatial sampling interval and improves traditional full-domain control schemes. Additionally, by utilizing variable rearrangement technique and constructing some Lyapunov-Krasovskii functionals, several passivity and synchronization criteria presented by LMIs are derived. Moreover, to further reduce the control cost, an adaptive spatial sampling control strategy is proposed and the adaptive synchronization is rigorously analyzed. Note that only the weighted union topology of all coupling layers is constrained to be connected, which relaxes the previous requirement on the strong connectivity of each coupling topology. The obtained results are confirmed by an illustrative example and are applied to image encryption and decryption.

Keywords: Coupled neural networks; Spatial sampling control; Passivity and synchronization; Multiweighted coupling; Adaptive control

1 Introduction

Over recent decades, coupled neural networks (CNNs) have attracted widespread attention due to their extensive applications in addressing complex practical problems, such as pattern recognition, associative memory and quadratic optimization [1–3]. However, most of these models are described by a single coupling, irrespective of the complexity of the relationships among real individuals. Although the use of a single coupling model offers the advantages of simplicity and ease of analysis, this simplification might

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lead to an incomplete understanding of the system’s complexity. For instance, in social networks, individuals communicate via multiple channels, including media platforms, emails and phone calls [4]. Similarly, the complexity of controlling COVID-19 has been intensified by its multiple transmission routes like person-to-person, fomite-to-person and environmental-to-person transmission [5]. To capture the richness of these interactions, network models incorporate multi-weighted couplings, which have important applications in biological network modeling, multi-modal data processing and economic systems [6–8].

Passivity, as a fundamental property of physical systems, characterizes the energy dissipation behavior via the system’s intrinsic energy storage dynamics and its external input-output relationships. Specifically, a passive system is a physical system with inherent dissipative components, in which the accumulated stored energy for any arbitrary time durations is less than or equal to the energy supplied through external inputs [9]. Typical examples include resistors, capacitors, and inductors in electrical circuits [10]. This property implies that the system cannot generate energy by itself but can only dissipate energy supplied by external sources. Consequently, passivity provides a powerful criterion for guaranteeing the intrinsic stability of systems [11]. Given its widespread utility in diverse domains such as mechanical systems and power networks, the passivity of CNNs has garnered growing interest, and a large number of excellent achievements have emerged [12–15]. Wang et al. [12] conducted a rigorous passivity analysis for CNNs with multi-weighted couplings, and passivity conditions were derived for systems with mismatched input-output dimensions. By employing the variable rearrange order technique, Lin and Liu addressed the passivity and control of multi-coupled dynamical networks [14].

Note that, the above considerable efforts are mainly devoted to CNNs modeled by ordinary differential systems, where the dynamic evolution merely depends on the time. However, these models neglect the spatial evolution of neurons and are unable to accurately explain the flow of neuronal information as well as the inter-layer information exchange among neurons in biological systems [16,17]. To accurately depict the spatiotemporal dynamics and interactions of neurons, the reaction-diffusion term is introduced into CNNs. Recently, passivity analysis and passivity-based control of reaction-diffusion neural networks (RDNNs) have been increasingly reported [18–21]. In [18], by establishing the definition of fixed-time passivity, some fixed-time passification conditions for memristive reaction-diffusion neural networks were obtained via nonlinear full-domain control. In practice, time delays are inevitable due to the limited bandwidth and the constrained transmission speed during information transmission [20]. In view of this, based on Lyapunov Krasovskii functionals (LKF) method, Huang et al. [21] explored the passivity

of multi-weighted RDNNs with time-varying delays via full-domain event-triggered controller. In [22], the input and output passivity of RDNNs with multi-proportional delays was addressed by constructing a LKF. More importantly, Wang et al. [23, 24] further extended the theory of passivity to investigate the synchronization of CNNs. Specifically, the synchronization problem of CNNs was reformulated as the stability analysis of an error system. When the external input is removed, the original CNNs achieves synchronization if the error system exhibits output-strict passivity. Since the concept of output-strict passivity was introduced, the problem of passivity-based control for synchronization of RDNNs has received considerable attention [25–27]. In [25], an adaptive pinning controller was designed to realize both passivity and synchronization in fractional spatiotemporal networks. Considering time delay and parameter uncertainties, Lin et al. studied passivity and synchronization of multiple-weighted RDNNs by designing an event-triggered controller in [26]. Note that, the aforementioned studies are exclusively concentrated on RDNNs characterized by the single time delay. However, in practical engineering and technical fields, a system may be simultaneously affected by multiple time-delay factors, such as constant delay, time-varying delay, distributed delay, and proportional delay [28–30]. Unfortunately, integrating these time delays into a unified system model and analyzing passivity and synchronization for RDNNs with multiple delays remain a challenging task.

It is important to highlight that the aforementioned passivity-based controllers require actuators to be distributed throughout the entire interior space domain. In contrast, spatial sampled-data control can achieve the desired control objective by distributing actuators at specific spatial points [31]. Owing to its significant advantages, including the ease of installation, cost-effectiveness and high efficiency, this approach has garnered considerable attention in both academic and practical fields [32–38]. Lu et al. [33] developed a kind of spacial sampled-data controller for H_∞ output synchronization of RDNNs with discrete and distributed delays, in which the actuators are placed at the midpoint of each sampling interval. In [34] and [35], a spatial sampled-data controller was presented to realize synchronization of RDNNs with and without mixed delays, and several conditions formed by linear matrix inequalities (LMI) were derived by constructing a LKF. Based on spatially averaged measurements and LKF technique, the dissipativity behaviors of delayed RDNNs were analysed via adaptive event-triggered sampled-data control in [36]. Unlike the LKF analysis technique, the cluster synchronization criteria for delayed genetic regulatory networks were derived based on Halanay’s inequality techniques under an intermittent space-dividing controller in [37]. Compared with the abundant achievements in the sampled-data synchronization of RDNNs, the passivity analysis of

delayed RDNNs under spatial sampling control has not been systematically and deeply explored. Given the diverse coupling structures and complex dynamic behaviors of delayed RDNNs, how to design efficient passivity-based sampling control strategies under spatial point measurements, and how to establish the intrinsic relationship between passivity and synchronization, are key scientific issues that need to be urgently addressed.

Inspired by the above discussion, the passivity and synchronization of delayed RDNNs with multi-weighted coupling are investigated in this article. Several sufficient conditions for passivity and synchronization are established by designing spatial sampling controllers. The main highlights of this article are summarized as follows.

(1) To capture the influence of different delays on the system and the richness of neural network interactions, a class of RDNNs with multi-weighted coupling is formulated which integrates discrete time delays [23, 36, 37], distributed delays [33, 34, 39], and proportional delays [22, 28].

(2) To realize passivity and synchronization, a spatial sampling controller distributed at the midpoint of each sampling interval is designed, which is more economical and convenient than the full-domain control presented in [25–27]. To further reduce the control gains determined by the LMI condition, an adaptive spatial sampling control strategy is developed to dynamically update the control gain and finally achieve asymptotical synchronization.

(3) By constructing some LKFs, several passivity and passivity-based synchronization criteria are derived for the RDNNs with mismatched input and output dimensions. In theoretical analysis, the requirement for strong connectivity of each coupling layer of communication topology, as considered in [6–8, 33], is relaxed to the connectivity about the union graph through the variable rearrangement technique.

Table 1: Notations and Definitions

Notations	Definitions
$\mathbb{N}^+, \overline{1}, \overline{\mathcal{N}}, \mathcal{R}_{\geq 0}$	$\mathbb{N}^+ = \{1, 2, \dots\}, \overline{1}, \overline{\mathcal{N}} = \{1, 2, \dots, \mathcal{N}\}, \mathcal{R}_{\geq 0} = [0, +\infty)$
$\mathcal{R}^n, \mathcal{R}^{n \times m}$	The space of all n -dimensional real vectors/ $n \times m$ real matrices
$\mathcal{M} > 0 (< 0)$	Matrix $\mathcal{M}$ is positive definite (negative definite)
$[\mathcal{M}]^s$	$[\mathcal{M}]^s = \frac{\mathcal{M}^T + \mathcal{M}}{2}$
$\ v\ _2$	$\ v\ _2 = \sqrt{v^T v}$ with $v \in \mathcal{R}^n$
$\ z(t, \cdot)\ _{[\underline{\alpha}, \overline{\alpha}]}$	$\ z(t, \cdot)\ _{[\underline{\alpha}, \overline{\alpha}]} = (\int_{\underline{\alpha}}^{\overline{\alpha}} z^T(t, \varsigma) z(t, \varsigma) d\varsigma)^{\frac{1}{2}}, t \in \mathcal{R}_{\geq 0}, z(t, \varsigma) \in \mathcal{R}^n$
$\mathbf{0}_n, \mathbf{1}_n$	$\mathbf{0}_n = (0, 0, \dots, 0)^T \in \mathcal{R}^n, \mathbf{1}_n = (1, 1, \dots, 1)^T \in \mathcal{R}^n$
$\mathbf{I}_n$	n -order identity matrix

2 Preparation and Problem Formation

Consider a kind of multi-weighted coupled reaction-diffusion neural networks with mixed time-varying delay, which is depicted as

$$\begin{aligned} \frac{\partial z_i(t, \varsigma)}{\partial t} = & \mathcal{D} \frac{\partial^2 z_i(t, \varsigma)}{\partial \varsigma^2} - \mathcal{A} z_i(t, \varsigma) + \mathcal{B}_1 f_1(z_i(t, \varsigma)) + \mathcal{B}_2 f_2(z_i(\varrho(t), \varsigma)) \\ & + \mathcal{B}_3 \int_{t-\iota(t)}^t f_3(z_i(v, \varsigma)) dv + \sum_{\eta=1}^{\sigma} \sum_{j=1}^{\mathcal{N}} c^\eta g_{ij}^\eta \Gamma^\eta z_j(t, \varsigma) \\ & + \omega_i(t, \varsigma) + u_i(t, \varsigma), \quad i \in \overline{1, \mathcal{N}}, \end{aligned} \quad (1)$$

where $t \in \mathcal{R}_{\geq 0}$, $\varsigma \in (\underline{\alpha}, \bar{\alpha})$, $z_i(t, \varsigma) = (z_i^1(t, \varsigma), z_i^2(t, \varsigma), \dots, z_i^n(t, \varsigma))^T \in \mathcal{R}^n$ is the state of the i -th neural network, $f_r(z_i(\cdot)) = (f_r^1(z_i^1(\cdot)), f_r^2(z_i^2(\cdot)), \dots, f_r^n(z_i^n(\cdot)))^T \in \mathcal{R}^n$ is the nonlinear activation function ($r = 1, 2, 3$), $\omega_i(t, \varsigma) \in \mathcal{R}^n$ is the external input, $u_i(t, \varsigma) \in \mathcal{R}^n$ is a spatial sampled-data controller, $\varrho(t)$ denotes the general time-varying delay, $\iota(t)$ represents the distributed time-varying delay, $\mathcal{D} = \text{diag}\{d_1, d_2, \dots, d_n\} > 0$ is the diffusion coefficient matrix, $\mathcal{A} = \text{diag}\{a_1, a_2, \dots, a_n\}$, $a_k > 0$ denotes the potential decay rate of the k -th neuron toward its resting state, $\mathcal{B}_r = (b_{ij}^r)_{n \times n}$ ($r \in \overline{1, 3}$) are the synaptic connection weights between neurons, $c^\eta > 0$ and $\Gamma^\eta = \text{diag}\{\gamma_1^\eta, \gamma_2^\eta, \dots, \gamma_n^\eta\} > 0$ are the coupling strength and the inner coupling matrix of the η -th coupling, respectively. $G^\eta = (g_{ij}^\eta)_{\mathcal{N} \times \mathcal{N}}$ represents the outer coupling matrix of the η -th coupling, where $g_{ij}^\eta > 0$ ($i \neq j$) if there is communication from node j to node i , otherwise, $g_{ij}^\eta = 0$, and $g_{ii}^\eta = -\sum_{j=1, j \neq i}^{\mathcal{N}} g_{ij}^\eta$.

Assumption 1. For any nonlinear function $f_r(\cdot) : \mathcal{R}^n \rightarrow \mathcal{R}^n$ ($r \in \overline{1, 3}$), there exists a positive constant L_r such that

$$\|f_r(z) - f_r(z^*)\|_2 \leq \sqrt{L_r} \|z - z^*\|_2, \quad z, z^* \in \mathcal{R}^n.$$

Assumption 2. There exist positive constants $\hat{\varrho}$, $\bar{\iota}$ and $\hat{\iota}$ such that

$$t \leq \varrho(t) + \bar{\varrho}, \quad 0 < \hat{\varrho} \leq \dot{\varrho}(t), \quad \iota(t) \leq \bar{\iota}.$$

The Dirichlet boundary condition and the initial state of system (1) are given as

$$\begin{cases} z_i(t, \underline{\alpha}) = z_i(t, \bar{\alpha}) = \mathbf{0}_n, & t \in [-\iota, +\infty), \\ z_i(t, \varsigma) = \phi_i(t, \varsigma), & (t, \varsigma) \in [-\iota, 0] \times (\underline{\alpha}, \bar{\alpha}), \end{cases} \quad (2)$$

where $\iota = \max\{-\varrho(0), \bar{\iota}\}$ with $\varrho(0) \leq 0$, and $\phi_i(t, \varsigma) = (\phi_i^1(t, \varsigma), \phi_i^2(t, \varsigma), \dots, \phi_i^n(t, \varsigma))^T \in \mathcal{R}^n$ is a continuous function.

Remark 1. The function $\varrho(t)$ represents a generalized form of time-delay. Specifically, if $\varrho(t) = t - h(t)$ with $0 \leq h(t) \leq h$, then $\varrho(t)$ reduces to a discrete time-delay as discussed

in [23, 36, 37]. If $\varrho(t) = pt$ with $0 < p < 1$, it simplifies to a proportional delay, which has been explored in [28].

From $z_i(t, \varsigma) = (z_i^1(t, \varsigma), z_i^2(t, \varsigma), \dots, z_i^n(t, \varsigma))^T$, system (1) is represented in the following scalar form

$$\begin{aligned} \frac{\partial z_i^k(t, \varsigma)}{\partial t} = & d_k \frac{\partial^2 z_i^k(t, \varsigma)}{\partial \varsigma^2} - a_k z_i^k(t, \varsigma) + \sum_{s=1}^n b_{ks}^1 f_1^s(z_i^s(t, \varsigma)) \\ & + \sum_{s=1}^n b_{ks}^2 f_2^s(z_i^s(\varrho(t), \varsigma)) + \sum_{s=1}^n b_{ks}^3 \int_{t-\iota(t)}^t f_3^s(z_i^s(v, \varsigma)) dv \\ & + \sum_{\eta=1}^{\sigma} \sum_{j=1}^{\mathcal{N}} c^{\eta} g_{ij}^{\eta} \gamma_k^{\eta} z_j^k(t, \varsigma) + \omega_i^k(t, \varsigma) + u_i^k(t, \varsigma). \end{aligned} \quad (3)$$

Define the weighted union matrices $\mathcal{G}^{(k)} \in R^{\mathcal{N} \times \mathcal{N}}$ as

$$\mathcal{G}^{(k)} = \sum_{\eta=1}^{\sigma} c^{\eta} \gamma_k^{\eta} G^{\eta}, \quad k \in \overline{1, n}.$$

Assumption 3. The communication topology of $\mathcal{G}^{(k)}$ is strong connected for $k \in \overline{1, n}$.

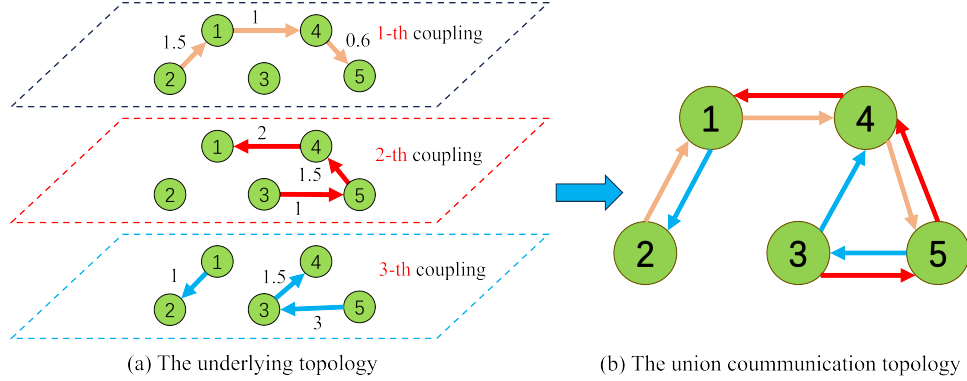


Fig. 1: The joint connectivity.

Remark 2. In the existing researches on RDNNs with multiweighted coupling [33], it is conventionally assumed that the communication topology of each layer's coupling is strongly connected. However, this study introduces a more relaxed condition. Specifically, only the union communication topology of all coupling layers is required to be connected in Assumption 3, as illustrated in Fig. 1, which is evidently less conservative.

Under Assumption 3, $\mathcal{G}^{(k)}$ has a simple zero eigenvalue and there exists a normalized vector $\xi^{(k)} = (\xi_1^k, \xi_2^k, \dots, \xi_{\mathcal{N}}^k)^T \in \mathcal{R}^{\mathcal{N}}$ with $\xi_m^k > 0$ ($m \in \overline{1, \mathcal{N}}$) satisfying $(\mathcal{G}^{(k)})^T \xi^{(k)} = \mathbf{0}_{\mathcal{N}}$ and $\sum_{i=1}^{\mathcal{N}} \xi_i^k = 1$. Based on this, define a virtual target as

$$\bar{z}^k(t, \varsigma) = \sum_{m=1}^{\mathcal{N}} \xi_m^k z_m^k(t, \varsigma), \quad (t, \varsigma) \in \mathcal{R}_{\geq 0} \times (\underline{\alpha}, \bar{\alpha}),$$

where $\bar{z}^k(t, \varsigma) \in \mathcal{R}$, $k \in \overline{1, n}$, and further denote $\varphi^k(t, \varsigma) = \sum_{m=1}^{\mathcal{N}} \xi_m^k \phi_m^k(t, \varsigma)$, $(t, \varsigma) \in [-\iota, 0] \times (\underline{\alpha}, \bar{\alpha})$.

Since $\mathcal{G}^{(k)T} \xi^{(k)} = \mathbf{0}_{\mathcal{N}}$, $\sum_{\eta=1}^{\sigma} \sum_{m=1}^{\mathcal{N}} c^\eta \gamma_k^\eta g_m^\eta \xi_m^k = 0$, $m \in \overline{1, \mathcal{N}}$, $k \in \overline{1, n}$, then

$$\begin{aligned} \frac{\partial \bar{z}^k(t, \varsigma)}{\partial t} &= d_k \frac{\partial^2 \bar{z}^k(t, \varsigma)}{\partial \varsigma^2} - a_k \bar{z}^k(t, \varsigma) + \sum_{m=1}^{\mathcal{N}} \xi_m^k \sum_{s=1}^n b_{ks}^1 f_1^s(z_m^s(t, \varsigma)) \\ &\quad + \sum_{m=1}^{\mathcal{N}} \xi_m^k \sum_{s=1}^n b_{ks}^2 f_2^s(z_m^s(\varrho(t), \varsigma)) + \sum_{m=1}^{\mathcal{N}} \xi_m^k \sum_{s=1}^n b_{ks}^3 \int_{t-\iota(t)}^t f_3^s(z_m^s(v, \varsigma)) dv \\ &\quad + \sum_{m=1}^{\mathcal{N}} \xi_m^k \omega_m^k(t, \varsigma) + \sum_{m=1}^{\mathcal{N}} \xi_m^k u_m^k(t, \varsigma). \end{aligned} \quad (4)$$

Denote the error $e_i^k(t, \varsigma) = z_i^k(t, \varsigma) - \bar{z}^k(t, \varsigma) \in \mathcal{R}$, it follows from systems (3) and (4) that

$$\begin{aligned} \frac{\partial e_i^k(t, \varsigma)}{\partial t} &= d_k \frac{\partial^2 e_i^k(t, \varsigma)}{\partial \varsigma^2} - a_k e_i^k(t, \varsigma) \\ &\quad + \sum_{s=1}^n b_{ks}^1 f_1^s(z_i^s(t, \varsigma)) - \sum_{m=1}^{\mathcal{N}} \xi_m^k \sum_{s=1}^n b_{ks}^1 f_1^s(z_m^s(t, \varsigma)) \\ &\quad + \sum_{s=1}^n b_{ks}^2 f_2^s(z_i^s(\varrho(t), \varsigma)) - \sum_{m=1}^{\mathcal{N}} \xi_m^k \sum_{s=1}^n b_{ks}^2 f_2^s(z_m^s(\varrho(t), \varsigma)) \\ &\quad + \sum_{s=1}^n b_{ks}^3 \int_{t-\iota(t)}^t f_3^s(z_i^s(v, \varsigma)) dv - \sum_{m=1}^{\mathcal{N}} \xi_m^k \sum_{s=1}^n b_{ks}^3 \int_{t-\iota(t)}^t f_3^s(z_m^s(v, \varsigma)) dv \\ &\quad + \omega_i^k(t, \varsigma) - \sum_{m=1}^{\mathcal{N}} \xi_m^k \omega_m^k(t, \varsigma) + u_i^k(t, \varsigma) - \sum_{m=1}^{\mathcal{N}} \xi_m^k u_m^k(t, \varsigma) \\ &\quad + \sum_{\eta=1}^{\sigma} \sum_{j=1}^{\mathcal{N}} c^\eta g_{ij}^\eta \gamma_k^\eta z_j^k(t, \varsigma), \quad i \in \overline{1, \mathcal{N}}, k \in \overline{1, n}. \end{aligned} \quad (5)$$

To explore the passivity of system (5), its output is given as

$$y_i(t, \varsigma) = P e_i(t, \varsigma) + Q \bar{\omega}_i(t, \varsigma), \quad i \in \overline{1, \mathcal{N}}, \quad (6)$$

where $y_i(t, \varsigma) = (y_i^1(t, \varsigma), \dots, y_i^p(t, \varsigma))^T \in \mathcal{R}^p$, $e_i(t, \varsigma) = (e_i^1(t, \varsigma), \dots, e_i^n(t, \varsigma))^T \in \mathcal{R}^n$, $\bar{\omega}_i(t, \varsigma) = (\bar{\omega}_i^1(t, \varsigma), \dots, \bar{\omega}_i^n(t, \varsigma))^T \in \mathcal{R}^n$, $\bar{\omega}_i^k(t, \varsigma) = \omega_i^k(t, \varsigma) - \sum_{m=1}^{\mathcal{N}} \xi_m^k \omega_m^k(t, \varsigma)$, and $P, Q \in \mathcal{R}^{p \times n}$.

Define

$$\hat{\omega}(t, \varsigma) = \left((\hat{\omega}^{(1)}(t, \varsigma))^T, (\hat{\omega}^{(2)}(t, \varsigma))^T, \dots, (\hat{\omega}^{(n)}(t, \varsigma))^T \right)^T \in \mathcal{R}^{n\mathcal{N}},$$

$$\begin{aligned}\hat{\omega}^{(k)}(t, \varsigma) &= (\bar{\omega}_1^k(t, \varsigma), \bar{\omega}_2^k(t, \varsigma), \dots, \bar{\omega}_{\mathcal{N}}^k(t, \varsigma))^T \in \mathcal{R}^{\mathcal{N}}, \\ \hat{y}(t, \varsigma) &= \left((y^{(1)}(t, \varsigma))^T, (y^{(2)}(t, \varsigma))^T, \dots, (y^{(1)}(t, \varsigma))^T \right)^T \in \mathcal{R}^{p\mathcal{N}}, \\ y^{(k)}(t, \varsigma) &= (y_1^k(t, \varsigma), y_2^k(t, \varsigma), \dots, y_{\mathcal{N}}^k(t, \varsigma))^T \in \mathcal{R}^{\mathcal{N}}.\end{aligned}$$

Definition 1 [25]. System (5) with input vector $\hat{\omega}(t, \varsigma) \in \mathcal{R}^{n\mathcal{N}}$ and output vector $\hat{y}(t, \varsigma) \in \mathcal{R}^{p\mathcal{N}}$ is said to be strictly passive if there exist a non-negative storage function $V(t)$ and matrices $\mathcal{H} \in \mathcal{R}^{p\mathcal{N} \times n\mathcal{N}}$, $0 < \mathcal{H}_1 \in \mathcal{R}^{n\mathcal{N} \times n\mathcal{N}}$, $0 < \mathcal{H}_2 \in \mathcal{R}^{p\mathcal{N} \times p\mathcal{N}}$ such that for any $t \in \mathcal{R}_{\geq 0}$,

$$\dot{V}(t) \leq \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{y}^T(t, \varsigma) \mathcal{H} \hat{\omega}(t, \varsigma) d\varsigma - \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{\omega}^T(t, \varsigma) \mathcal{H}_1 \hat{\omega}(t, \varsigma) d\varsigma - \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{y}^T(t, \varsigma) \mathcal{H}_2 \hat{y}(t, \varsigma) d\varsigma.$$

Furthermore, system (5) is said to be passive if $\mathcal{H}_1 = \mathbf{0}_{n\mathcal{N} \times n\mathcal{N}}$ and $\mathcal{H}_2 = \mathbf{0}_{p\mathcal{N} \times p\mathcal{N}}$. System (5) is input-strictly passive if $\mathcal{H}_1 > 0$ and $\mathcal{H}_2 = \mathbf{0}_{p\mathcal{N} \times p\mathcal{N}}$, system (5) is output-strictly passive if $\mathcal{H}_2 > 0$ and $\mathcal{H}_1 = \mathbf{0}_{n\mathcal{N} \times n\mathcal{N}}$.

Lemma 1 [17]. For any scalar $\varepsilon > 0$, vectors x and y , then $2x^T y \leq \varepsilon x^T x + \varepsilon^{-1} y^T y$.

Lemma 2 [40]. For any scalars $x_2 > x_1 > 0$, and a square-integrable vector function $z : [x_1, x_2] \rightarrow \mathcal{R}^n$,

$$\left(\int_{x_1}^{x_2} z(v) dv \right)^T \left(\int_{x_1}^{x_2} z(v) dv \right) \leq (x_2 - x_1) \int_{x_1}^{x_2} z^T(v) z(v) dv.$$

Lemma 3 [41]. Assume that $z : [\underline{\alpha}, \bar{\alpha}] \rightarrow \mathcal{R}^n$ be a square-integrable vector function with $z(a) = 0$ or $z(b) = 0$ and $0 < \mathcal{M} \in \mathcal{R}^{n \times n}$, then

$$\int_{\underline{\alpha}}^{\bar{\alpha}} z(\xi)^T \mathcal{M} z(\xi) d\xi \leq \frac{4(\bar{\alpha} - \underline{\alpha})^2}{\pi^2} \int_{\underline{\alpha}}^{\bar{\alpha}} \left(\frac{dz}{d\xi} \right)^T \mathcal{M} \left(\frac{dz}{d\xi} \right) d\xi.$$

Moreover, if $z(a) = z(b) = 0$, then

$$\int_{\underline{\alpha}}^{\bar{\alpha}} z(\xi)^T \mathcal{M} z(\xi) d\xi \leq \frac{(\bar{\alpha} - \underline{\alpha})^2}{\pi^2} \int_{\underline{\alpha}}^{\bar{\alpha}} \left(\frac{dz}{d\xi} \right)^T \mathcal{M} \left(\frac{dz}{d\xi} \right) d\xi.$$

3 Main results

In this section, the passivity and synchronization for RDNNs are investigated by designing the spatial sampling controller.

For the convenience of subsequent analysis, denote $\mathcal{P} = P \otimes \mathbf{I}_{\mathcal{N}}$, $\mathcal{Q} = Q \otimes \mathbf{I}_{\mathcal{N}}$, $\Lambda^{(k)} = \text{diag}\{\xi_1^k, \xi_2^k, \dots, \xi_{\mathcal{N}}^k\}$, $\Lambda = \text{diag}\{\Lambda^{(1)}, \Lambda^{(2)}, \dots, \Lambda^{(n)}\} \in \mathcal{R}^{n\mathcal{N} \times n\mathcal{N}}$, $\rho = 1 - \frac{\varepsilon_4}{2} - \frac{\delta^2}{2\varepsilon_4(\bar{\alpha} - \underline{\alpha})^2}$, $\mathcal{G} = \text{diag}\{\mathcal{G}^{(1)}, \mathcal{G}^{(2)}, \dots, \mathcal{G}^{(n)}\} \in \mathcal{R}^{n\mathcal{N} \times n\mathcal{N}}$, and

$$\Psi = \Lambda \mathcal{G} - \Lambda \left\{ \left[\frac{\pi^2}{(\bar{\alpha} - \underline{\alpha})^2} \mathcal{D} + \mathcal{A} - \sum_{r=1}^3 \frac{\mathcal{B}_r \mathcal{B}_r^T}{2\varepsilon_r} - \left(\frac{\varepsilon_1 L_1}{2} + \frac{\varepsilon_2 L_2}{2\hat{\rho}} + \frac{\varepsilon_3 L_3 \bar{t}^2}{2} \right) \mathbf{I}_n \right] \otimes \mathbf{I}_{\mathcal{N}} \right\} - \rho \Lambda \varpi.$$

3.1 Passivity for RDNNs

First of all, we consider the passivity of system (5) by designing a kind of sampled-data control strategy, the control architecture is shown in Fig. 2. A set of fixed points ς_p is placed such that $\underline{\alpha} = \varsigma_0 < \varsigma_1 < \dots < \varsigma_{m-1} < \varsigma_m = \bar{\alpha}$, $[\varsigma_p, \varsigma_{p+1})$ represents the sampling interval and satisfies $\sup_{p \in \overline{0, m-1}} \{\varsigma_{p+1} - \varsigma_p\} = \delta$, $\bar{\varsigma}_p = \frac{\varsigma_p + \varsigma_{p+1}}{2}$ is the sampling point.

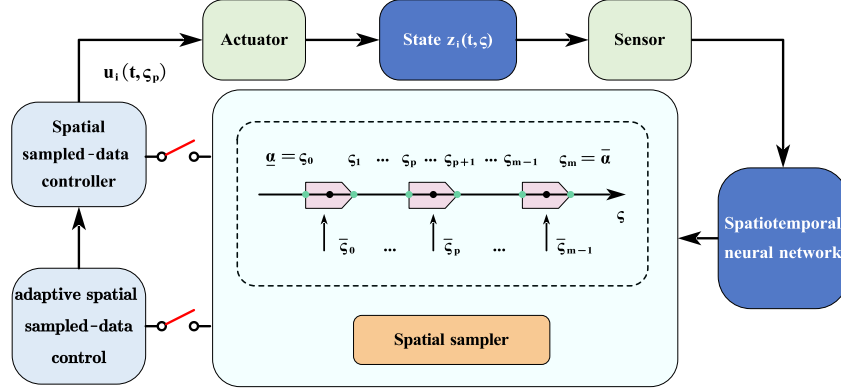


Fig. 2: Spatial sampling control architecture.

To realize the passivity of system (5), m sensors are placed on the midpoint of each sample subintervals $[\varsigma_p, \varsigma_{p+1})$ and the controller is designed as

$$u_i(t, \varsigma) = -\varpi_i e_i(t, \bar{\varsigma}_p), \quad \varsigma \in [\varsigma_p, \varsigma_{p+1}), \quad p \in \overline{0, m-1}, \quad (7)$$

where $t \in \mathcal{R}_{\geq 0}$, $\varpi = \mathbf{I}_n \otimes \text{diag}\{\varpi_1, \varpi_2, \dots, \varpi_N\}$, ϖ_i is the control gain.

Theorem 1. Under Assumptions 1-3 and spatial sampling controller (7), system (5) is strictly passive if there exist scalars $\varepsilon_i > 0$ ($i \in \overline{1, 4}$), matrix $\mathcal{H}$, positive-definite matrices $\mathcal{H}_1$ and $\mathcal{H}_2$ such that

$$C_1 : \quad \Xi = \begin{pmatrix} \Upsilon & \Sigma \\ \Sigma^T & \Delta \end{pmatrix} \leq 0, \\ C_2 : \quad \frac{\delta^2}{2\varepsilon_4\pi^2}\varpi - \mathcal{D} \otimes \mathbf{I}_N \leq 0,$$

where $\Upsilon = [\Psi - \mathcal{P}^T \mathcal{H}_2 \mathcal{P}]^s$, $\Sigma = \frac{1}{2}\Lambda + \mathcal{P}^T[\mathcal{H}_2]^s \mathcal{Q} - \frac{1}{2}\mathcal{P}^T \mathcal{H}$, $\Delta = \mathcal{Q}^T[\mathcal{H}_2]^s \mathcal{Q} + [\mathcal{H}_1]^s - [\mathcal{Q}^T \mathcal{H}]^s$.

Proof. Pick a Lyapunov functional

$$V(t) = \mathcal{U}(t) + \eta_1 \int_{\varrho(t)}^t \mathcal{U}(v) dv + \eta_2 \int_{-\bar{\varepsilon}}^0 \left[\int_{t+\theta}^t \mathcal{U}(v) dv \right] d\theta,$$

and

$$\mathcal{U}(t) = \frac{1}{2} \sum_{k=1}^n \int_{\underline{\alpha}}^{\bar{\alpha}} \left(e^{(k)}(t, \varsigma) \right)^T \Lambda^{(k)} e^{(k)}(t, \varsigma) d\varsigma,$$

where $t \in \mathcal{R}_{\geq 0}$, $\eta_1 = \frac{\varepsilon_2 L_2}{\bar{\varrho}}$, $\eta_2 = \varepsilon_3 L_3 \bar{\iota}$, $e^{(k)} = (e_1^k(t, \varsigma), \dots, e_{\mathcal{N}}^k(t, \varsigma))^T$, $k \in \overline{1, n}$, $\hat{e}(t, \varsigma) = \left((e^{(1)}(t, \varsigma))^T, (e^{(2)}(t, \varsigma))^T, \dots, (e^{(n)}(t, \varsigma))^T \right)^T \in \mathcal{R}^{n\mathcal{N}}$.

Take the derivative to $\mathcal{U}(t)$ along with error system (5),

$$\begin{aligned}
\dot{\mathcal{U}}(t) = & \sum_{i=1}^{\mathcal{N}} \sum_{k=1}^n \int_{\underline{\alpha}}^{\bar{\alpha}} \xi_i^k e_i^k(t, \varsigma) \left[d_k \frac{\partial^2 e_i^k(t, \varsigma^2)}{\partial \varsigma^2} - a_k e_i^k(t, \varsigma) \right. \\
& + \sum_{s=1}^n b_{ks}^1 f_1^s(z_i^s(t, \varsigma)) - \sum_{s=1}^n b_{ks}^1 f_1^s(\bar{z}^s(t, \varsigma)) \\
& + \sum_{s=1}^n b_{ks}^1 f_1^s(\bar{z}^s(t, \varsigma)) - \sum_{m=1}^{\mathcal{N}} \xi_m^k \sum_{s=1}^n b_{ks}^1 f_1^s(z_m^s(t, \varsigma)) \\
& + \sum_{s=1}^n b_{ks}^2 f_2^s(z_i^s(\varrho(t), \varsigma)) - \sum_{s=1}^n b_{ks}^2 f_2^s(\bar{z}^s(\varrho(t), \varsigma)) \\
& + \sum_{s=1}^n b_{ks}^2 f_2^s(\bar{z}_i^s(\varrho(t), \varsigma)) - \sum_{m=1}^{\mathcal{N}} \xi_m^k \sum_{s=1}^n b_{ks}^2 f_2^s(z_m^s(\varrho(t), \varsigma)) \\
& + \sum_{s=1}^n b_{ks}^3 \int_{t-\iota(t)}^t f_3^s(z_i^s(v, \varsigma)) dv - \sum_{s=1}^n b_{ks}^3 \int_{t-\iota(t)}^t f_3^s(\bar{z}^s(v, \varsigma)) dv \\
& + \sum_{s=1}^n b_{ks}^3 \int_{t-\iota(t)}^t f_3^s(\bar{z}^s(v, \varsigma)) dv - \sum_{m=1}^{\mathcal{N}} \xi_m^k \sum_{s=1}^n b_{ks}^3 \int_{t-\iota(t)}^t f_3^s(z_i^s(v, \varsigma)) dv \\
& \left. + \sum_{\eta=1}^{\sigma} \sum_{j=1}^{\mathcal{N}} c^{\eta} g_{ij}^{\eta} \gamma_k^{\eta} z_j^k(t, \varsigma) + \bar{\omega}_i^k(t, \varsigma) + u_i^k(t, \varsigma) - \sum_{m=1}^{\mathcal{N}} \xi_m^k u_m^k(t, \varsigma) \right] d\varsigma. \quad (8)
\end{aligned}$$

According to the Dirichlet boundary condition and utilizing integration by parts,

$$\begin{aligned}
& \sum_{i=1}^{\mathcal{N}} \sum_{k=1}^n \xi_i^k \int_{\underline{\alpha}}^{\bar{\alpha}} e_i^k(t, \varsigma) d_k \frac{\partial^2 e_i^k(t, \varsigma)}{\partial \varsigma^2} d\varsigma \\
= & \sum_{i=1}^{\mathcal{N}} \sum_{k=1}^n e_i^k \xi_i^k d_k \frac{\partial e_i^k(t, \varsigma)}{\partial \varsigma} \Big|_{\underline{\alpha}}^{\bar{\alpha}} - \sum_{i=1}^{\mathcal{N}} \sum_{k=1}^n \int_{\underline{\alpha}}^{\bar{\alpha}} \frac{\partial e_i^k(t, \varsigma)}{\partial \varsigma} \xi_i^k d_k \frac{\partial e_i^k(t, \varsigma)}{\partial \varsigma} d\varsigma \\
= & - \sum_{k=1}^n \int_{\underline{\alpha}}^{\bar{\alpha}} d_k \left(\frac{\partial e^{(k)}(t, \varsigma)}{\partial \varsigma} \right)^T \Lambda^{(k)} \left(\frac{\partial e^{(k)}(t, \varsigma)}{\partial \varsigma} \right) d\varsigma \\
= & - \int_{\underline{\alpha}}^{\bar{\alpha}} \left(\frac{\partial \hat{e}(t, \varsigma)}{\partial \varsigma} \right)^T \Lambda(\mathcal{D} \otimes \mathbf{I}_{\mathcal{N}}) \left(\frac{\partial \hat{e}(t, \varsigma)}{\partial \varsigma} \right) d\varsigma. \quad (9)
\end{aligned}$$

It is noted that

$$\sum_{i=1}^{\mathcal{N}} \xi_i^k e_i^k(t, \varsigma) = \sum_{i=1}^{\mathcal{N}} \xi_i^k \left[z_i^k(t, \varsigma) - \sum_{m=1}^{\mathcal{N}} \xi_m^k z_m^k(t, \varsigma) \right] = 0,$$

which shows that

$$\begin{aligned}
& \sum_{i=1}^{\mathcal{N}} \sum_{k=1}^n \xi_i^k e_i^k(t, \varsigma) \left[\sum_{s=1}^n b_{ks}^1 f_1^s(\bar{z}^s(t, \varsigma)) - \sum_{m=1}^{\mathcal{N}} \xi_m^k \sum_{s=1}^n b_{ks}^1 f_1^s(z_m^s(t, \varsigma)) \right] = 0, \\
& \sum_{i=1}^{\mathcal{N}} \sum_{k=1}^n \xi_i^k e_i^k(t, \varsigma) \left[\sum_{s=1}^n b_{ks}^2 f_2^s(\bar{z}^s(\varrho(t), \varsigma)) - \sum_{m=1}^{\mathcal{N}} \xi_m^k \sum_{s=1}^n b_{ks}^2 f_2^s(z_m^s(\varrho(t), \varsigma)) \right] = 0, \\
& \sum_{i=1}^{\mathcal{N}} \sum_{k=1}^n \xi_i^k e_i^k(t, \varsigma) \int_{t-\iota(t)}^t \left[\sum_{s=1}^n b_{ks}^3 f_3^s(\bar{z}^s(v, \varsigma)) - \sum_{m=1}^{\mathcal{N}} \xi_m^k \sum_{s=1}^n b_{ks}^3 f_3^s(z_m^s(v, \varsigma)) \right] dv = 0, \\
& \sum_{i=1}^{\mathcal{N}} \sum_{k=1}^n \xi_i^k e_i^k(t, \varsigma) \left[\sum_{m=1}^{\mathcal{N}} \xi_m^k u_m^k(t, \varsigma) \right] = 0.
\end{aligned}$$

Define

$$\begin{aligned}
\tilde{e}_i(t, \varsigma) &= \text{diag}\{\sqrt{\xi_i^n}, \dots, \sqrt{\xi_i^n}\} e_i(t, \varsigma) \in \mathcal{R}^n, \\
\tilde{F}_r(e_i(\cdot, \varsigma)) &= \text{diag}\{\sqrt{\xi_i^n}, \dots, \sqrt{\xi_i^n}\} (f_r(z_i(\cdot, \varsigma)) - f_r(\bar{z}(\cdot, \varsigma))) \in \mathcal{R}^n.
\end{aligned}$$

From Assumption 1, Lemmas 1 and 2, there exist some positive scalars ε_i ($i \in \overline{1, 3}$) such that

$$\begin{aligned}
& \sum_{k=1}^n e_i^k(t, \varsigma) \xi_i^k \sum_{s=1}^n b_{ks}^1 \left[f_1^s(z_i^s(t, \varsigma)) - f_1^s(\bar{z}^s(t, \varsigma)) \right] \\
& + \sum_{k=1}^n e_i^k(t, \varsigma) \xi_i^k \sum_{s=1}^n b_{ks}^2 \left[f_2^s(z_i^s(\varrho(t), \varsigma)) - f_2^s(\bar{z}^s(\varrho(t), \varsigma)) \right] \\
& + \sum_{k=1}^n e_i^k(t, \varsigma) \xi_i^k \sum_{s=1}^n b_{ks}^3 \int_{t-\iota(t)}^t \left[f_3^s(z_i^s(v, \varsigma)) - f_3^s(\bar{z}^s(v, \varsigma)) \right] dv \\
& \leq \frac{1}{2\varepsilon_1} \sum_{i=1}^{\mathcal{N}} \tilde{e}_i^T(t, \varsigma) \mathcal{B}_1 \mathcal{B}_1^T \tilde{e}_i(t, \varsigma) + \frac{\varepsilon_1 L_1}{2} \sum_{i=1}^{\mathcal{N}} \tilde{e}_i^T(t, \varsigma) \tilde{e}_i(t, \varsigma) \\
& + \frac{1}{2\varepsilon_2} \sum_{i=1}^{\mathcal{N}} \tilde{e}_i^T(t, \varsigma) \mathcal{B}_2 \mathcal{B}_2^T \tilde{e}_i(t, \varsigma) + \frac{\varepsilon_2 L_2}{2} \sum_{i=1}^{\mathcal{N}} \tilde{e}_i^T(\varrho(t), \varsigma) \tilde{e}_i(\varrho(t), \varsigma) \\
& + \frac{1}{2\varepsilon_3} \sum_{i=1}^{\mathcal{N}} \tilde{e}_i^T(t, \varsigma) \mathcal{B}_3 \mathcal{B}_3^T \tilde{e}_i(t, \varsigma) + \frac{\varepsilon_3 L_3 \bar{\iota}}{2} \sum_{i=1}^{\mathcal{N}} \int_{t-\iota(t)}^t \tilde{e}_i^T(v, \varsigma) \tilde{e}_i(v, \varsigma) dv \\
& \leq \frac{1}{2} \hat{e}^T(t, \varsigma) \Lambda \left(\sum_{r=1}^3 \frac{1}{\varepsilon_r} \mathcal{B}_r \mathcal{B}_r^T \otimes \mathbf{I}_{\mathcal{N}} \right) \hat{e}(t, \varsigma) + \frac{\varepsilon_1 L_1}{2} \hat{e}^T(t, \varsigma) \Lambda \hat{e}(t, \varsigma) \\
& + \frac{\varepsilon_2 L_2}{2} \hat{e}^T(\varrho(t), \varsigma) \Lambda \hat{e}(\varrho(t), \varsigma) + \frac{\varepsilon_3 L_3 \bar{\iota}}{2} \int_{t-\iota(t)}^t \hat{e}^T(v, \varsigma) \Lambda \hat{e}(v, \varsigma) dv. \tag{10}
\end{aligned}$$

Note that $e_i^k(t, \bar{\varsigma}_p) = e_i^k(t, \varsigma) - \int_{\bar{\varsigma}_p}^{\varsigma} \frac{\partial e_i^k(t, s)}{\partial s} ds$, $\varsigma_{p+1} - \bar{\varsigma}_p \leq \frac{\delta}{2}$ and $\bar{\varsigma}_p - \varsigma_p \leq \frac{\delta}{2}$, then

$$\begin{aligned}
& \sum_{i=1}^{\mathcal{N}} \sum_{k=1}^n \int_{\underline{\alpha}}^{\bar{\alpha}} e_i^k(t, \varsigma) \xi_i^k u_i^k(t, \varsigma) d\varsigma \\
&= - \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \Lambda \varpi \hat{e}(t, \varsigma) d\varsigma + \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \Lambda \varpi \left(\int_{\bar{\varsigma}_p}^{\varsigma} \frac{\partial \hat{e}(t, s)}{\partial s} ds \right) d\varsigma \\
&\leq - \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \Lambda \varpi \hat{e}(t, \varsigma) d\varsigma + \sum_{p=0}^{m-1} \frac{\varepsilon_4}{2} \int_{x_p}^{x_{p+1}} \hat{e}^T(t, \varsigma) \Lambda \varpi \hat{e}(t, \varsigma) d\varsigma \\
&\quad + \frac{1}{2\varepsilon_4} \sum_{p=0}^{m-1} \int_{\varsigma_p}^{\varsigma_{p+1}} \left(\hat{e}(t, \varsigma) - \hat{e}(t, \bar{\varsigma}_p) \right)^T \Lambda \varpi \left(\hat{e}(t, \varsigma) - \hat{e}(t, \bar{\varsigma}_p) \right) d\varsigma \\
&= - \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \Lambda \varpi \hat{e}(t, \varsigma) d\varsigma + \frac{\varepsilon_4}{2} \sum_{p=0}^{m-1} \int_{\varsigma_p}^{\varsigma_{p+1}} \hat{e}^T(t, \varsigma) \Lambda \varpi \hat{e}(t, \varsigma) d\varsigma \\
&\quad + \frac{1}{2\varepsilon_4} \sum_{p=0}^{m-1} \left[\int_{\varsigma_p}^{\bar{\varsigma}_p} \left(\hat{e}(t, \varsigma) - \hat{e}(t, \bar{\varsigma}_p) \right)^T \Lambda \varpi \left(\hat{e}(t, \varsigma) - \hat{e}(t, \bar{\varsigma}_p) \right) d\varsigma \right. \\
&\quad \left. + \int_{\bar{\varsigma}_p}^{\varsigma_{p+1}} \left(\hat{e}(t, \varsigma) - \hat{e}(t, \bar{\varsigma}_p) \right)^T \Lambda \varpi \left(\hat{e}(t, \varsigma) - \hat{e}(t, \bar{\varsigma}_p) \right) d\varsigma \right] \\
&\leq - \left(1 - \frac{\varepsilon_4}{2}\right) \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \Lambda \varpi \hat{e}(t, \varsigma) d\varsigma \\
&\quad + \frac{1}{2\varepsilon_4} \sum_{p=0}^{m-1} \left[\frac{4(\bar{\varsigma}_p - \varsigma_p)^2}{\pi^2} \int_{\varsigma_p}^{\bar{\varsigma}_p} \left(\frac{\partial \hat{e}(t, \varsigma)}{\partial \varsigma} \right)^T \Lambda \varpi \frac{\partial \hat{e}(t, \varsigma)}{\partial \varsigma} d\varsigma \right. \\
&\quad \left. + \frac{4(\varsigma_{p+1} - \bar{\varsigma}_p)^2}{\pi^2} \int_{\bar{\varsigma}_p}^{\varsigma_{p+1}} \left(\frac{\partial \hat{e}(t, \varsigma)}{\partial \varsigma} \right)^T \Lambda \varpi \frac{\partial \hat{e}(t, \varsigma)}{\partial \varsigma} d\varsigma \right] \\
&\leq - \left(1 - \frac{\varepsilon_4}{2}\right) \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \Lambda \varpi \hat{e}(t, \varsigma) d\varsigma \\
&\quad + \frac{\delta^2}{2\varepsilon_4 \pi^2} \int_{\underline{\alpha}}^{\bar{\alpha}} \left(\frac{\partial \hat{e}(t, \varsigma)}{\partial \varsigma} \right)^T \Lambda \varpi \frac{\partial \hat{e}(t, \varsigma)}{\partial \varsigma} d\varsigma. \tag{11}
\end{aligned}$$

According to the condition C_2 , equation (9), inequality (11) and Lemma 3,

$$\begin{aligned}
& - \int_{\underline{\alpha}}^{\bar{\alpha}} \frac{\partial \hat{e}^T(t, \varsigma)}{\partial \varsigma} \Lambda \left[\mathcal{D} \otimes \mathbf{I}_{\mathcal{N}} - \frac{\delta^2}{2\varepsilon_4 \pi^2} \varpi \right] \frac{\partial \hat{e}(t, \varsigma)}{\partial \varsigma} d\varsigma \\
&\leq - \frac{1}{(\bar{\alpha} - \underline{\alpha})^2} \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \Lambda \left[\pi^2 \mathcal{D} \otimes \mathbf{I}_{\mathcal{N}} - \frac{\delta^2}{2\varepsilon_4} \varpi \right] \hat{e}(t, \varsigma) d\varsigma. \tag{12}
\end{aligned}$$

Substituting the formulas (9)-(12) into the equation (8) yields that

$$\begin{aligned}
\dot{\mathcal{U}}(t) \leq & \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \left\{ [\Lambda \mathcal{G}]^s - \Lambda \left[\left(\frac{\pi^2}{(\bar{\alpha} - \underline{\alpha})^2} \mathcal{D} + \mathcal{A} - \sum_{r=1}^3 \frac{\mathcal{B}_r \mathcal{B}_r^T}{2\varepsilon_r} - \frac{\varepsilon_1 L_1}{2} \mathbf{I}_n \right) \otimes \mathbf{I}_{\mathcal{N}} \right] \right. \\
& - \left(1 - \frac{\varepsilon_4}{2} - \frac{\delta^2}{2\varepsilon_4(\bar{\alpha} - \underline{\alpha})^2} \right) \Lambda \varpi \left. \right\} \hat{e}(t, \varsigma) d\varsigma + \frac{\varepsilon_2 L_2}{2} \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(\varrho(t), \varsigma) \Lambda \hat{e}(\varrho(t), \varsigma) d\varsigma \\
& + \frac{\varepsilon_3 L_3 \bar{\ell}}{2} \int_{\underline{\alpha}}^{\bar{\alpha}} \int_{t-\iota(t)}^t \hat{e}^T(v, \varsigma) \Lambda \hat{e}(v, \varsigma) dv d\varsigma + \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \Lambda \hat{\omega}(t, \varsigma) d\varsigma. \tag{13}
\end{aligned}$$

Combining inequality (13) and Assumption 2,

$$\begin{aligned}
\dot{V}(t) = & \dot{\mathcal{U}}(t) + \eta_1 \mathcal{U}(t) - \eta_1 \dot{\varrho}(t) \mathcal{U}(\varrho(t)) + \eta_2 \left(\bar{\mathcal{U}}(t) - \int_{t-\bar{\ell}}^t \mathcal{U}(v) dv \right) \\
\leq & \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \left\{ [\Lambda \mathcal{G}]^s - \Lambda \left[\left(\frac{\pi^2}{(\bar{\alpha} - \underline{\alpha})^2} \mathcal{D} + \mathcal{A} - \frac{1}{2} \sum_{r=1}^3 \frac{1}{\varepsilon_r} \mathcal{B}_r \mathcal{B}_r^T \right. \right. \right. \\
& - \left. \left. \frac{\varepsilon_1 L_1 + \eta_1 + \eta_2 \bar{\ell}}{2} \mathbf{I}_n \right) \otimes \mathbf{I}_{\mathcal{N}} \right] - \left(1 - \frac{\varepsilon_4}{2} - \frac{\delta^2}{2\varepsilon_4(\bar{\alpha} - \underline{\alpha})^2} \right) \Lambda \varpi \left. \right\} \hat{e}(t, \varsigma) d\varsigma \\
& + \left(\frac{\varepsilon_2 L_2}{2} - \frac{\eta_1 \dot{\varrho}(t)}{2} \right) \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(\varrho(t), \varsigma) \Lambda \hat{e}(\varrho(t), \varsigma) d\varsigma \\
& + \left(\frac{\varepsilon_3 L_3 \bar{\ell}}{2} - \frac{\eta_2}{2} \right) \int_{\underline{\alpha}}^{\bar{\alpha}} \int_{t-\iota(t)}^t \hat{e}^T(v, \varsigma) \Lambda \hat{e}(v, \varsigma) dv d\varsigma \\
& + \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \Lambda \hat{\omega}(t, \varsigma) d\varsigma \\
\leq & \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \Psi \hat{e}(t, \varsigma) d\varsigma + \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \Lambda \hat{\omega}(t, \varsigma) d\varsigma.
\end{aligned}$$

Furthermore, according to Definition 1, one obtains

$$\begin{aligned}
\dot{V}(t) = & \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{y}^T(t, \varsigma) \mathcal{H} \hat{\omega}(t, \varsigma) d\varsigma + \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{\omega}^T(t, \varsigma) \mathcal{H}_1 \hat{\omega}(t, \varsigma) d\varsigma + \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{y}^T(t, \varsigma) \mathcal{H}_2 \hat{y}(t, \varsigma) d\varsigma \\
\leq & \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \left[\Psi - \mathcal{P}^T \mathcal{H}_2 \mathcal{P} \right]^s \hat{e}(t, \varsigma) d\varsigma \\
& + \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \left(\frac{1}{2} \Lambda + \mathcal{P}^T [\mathcal{H}_2]^s \mathcal{Q} - \frac{1}{2} \mathcal{P}^T \mathcal{H} \right) \hat{\omega}(t, \varsigma) d\varsigma \\
& + \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{\omega}^T(t, \varsigma) \left(\frac{1}{2} \Lambda + \mathcal{P}^T [\mathcal{H}_2]^s \mathcal{Q} - \frac{1}{2} \mathcal{H}^T \mathcal{P} \right) \hat{e}(t, \varsigma) d\varsigma \\
& + \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{\omega}^T(t, \varsigma) d\varsigma \left(\mathcal{Q}^T [\mathcal{H}_2]^s \mathcal{Q} + [\mathcal{H}_1]^s - [\mathcal{Q}^T \mathcal{H}]^s \right) \hat{\omega}(t, \varsigma) d\varsigma
\end{aligned}$$

$$= \int_{\underline{\alpha}}^{\bar{\alpha}} \mathbf{E}^T(t, \varsigma) \Xi \mathbf{E}(t, \varsigma) d\varsigma, \quad (14)$$

where $\mathbf{E}(t, \varsigma) = (\hat{e}^T(t, \varsigma), \hat{\omega}^T(t, \varsigma))^T$.

By combining the condition C_1 and inequality (14),

$$\dot{V}(t) \leq \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{y}^T(t, \varsigma) \mathcal{H} \hat{\omega}(t, \varsigma) d\varsigma - \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{\omega}^T(t, \varsigma) \mathcal{H}_1 \hat{\omega}(t, \varsigma) d\varsigma - \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{y}^T(t, \varsigma) \mathcal{H}_2 \hat{y}(t, \varsigma) d\varsigma.$$

The proof is finished. $\square$

Corollary 1. Under Assumptions 1-3 and spatial sampled-data controller (7), the error system (5) is input-strictly passive if there exist scalars $\varepsilon_i > 0$ ($i \in \overline{1, 4}$) and matrices $\mathcal{H} \in \mathcal{R}^{p\mathcal{N} \times n\mathcal{N}}$ and $0 < \mathcal{H}_1 \in \mathcal{R}^{n\mathcal{N} \times n\mathcal{N}}$ such that

$$\tilde{\Xi} = \begin{pmatrix} [\Psi]^s & \tilde{\Sigma} \\ \tilde{\Sigma}^T & \tilde{\Delta} \end{pmatrix} \leq 0, \quad \frac{\delta^2}{2\varepsilon_4\pi^2} \varpi - \mathcal{D} \otimes \mathbf{I}_{\mathcal{N}} \leq 0,$$

where $\tilde{\Sigma} = \frac{1}{2}\Lambda - \frac{1}{2}\mathcal{P}^T\mathcal{H}$, $\tilde{\Delta} = [\mathcal{H}_1]^s - [\mathcal{Q}^T\mathcal{H}]^s$.

Corollary 2. Under Assumptions 1-3 and spatial sampled-data controller (7), the error system (5) is output-strictly passive if there exist scalars $\varepsilon_i > 0$ ($i \in \overline{1, 4}$) and matrices $\mathcal{H} \in \mathcal{R}^{p\mathcal{N} \times n\mathcal{N}}$ and $0 < \mathcal{H}_2 \in \mathcal{R}^{p\mathcal{N} \times p\mathcal{N}}$ such that

$$\hat{\Xi} = \begin{pmatrix} \Upsilon & \Sigma \\ \Sigma^T & \hat{\Delta} \end{pmatrix} \leq 0, \quad \frac{\delta^2}{2\varepsilon_4\pi^2} \varpi - \mathcal{D} \otimes \mathbf{I}_{\mathcal{N}} \leq 0,$$

where $\hat{\Delta} = \mathcal{Q}^T[\mathcal{H}_2]^s\mathcal{Q} - [\mathcal{Q}^T\mathcal{H}]^s$.

Remark 3. The established conditions in Theorem 1 depend on the control parameters ϖ_i ($i \in \overline{1, \mathcal{N}}$), the matrices $\mathcal{H}$, $\mathcal{H}_1$, $\mathcal{H}_2$, and the free parameters $\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4$. By Schur complement Lemma [42], $\Xi \leq 0$ is equivalent to

$$\Delta < 0 \quad \text{and} \quad R - \Sigma\Delta^{-1}\Sigma^T \leq 0.$$

When other parameters are fixed, an increase in the control gain ϖ_i leads to a higher probability of satisfying the condition $\Xi \leq 0$, which implies that the passivity of the error system (5) can be easily ensured by implementing control with large gains. However, it is imperative to avoid large control actions in practice. Therefore, some appropriate free parameters ε_r ($r \in \overline{1, 4}$) can be selected as a tradeoff between convergence performance and control gains. In addition, a decrease in the upper bound of sampling interval δ facilitates the fulfillment of condition (C_2). The associated parameter selection strategy is presented in Algorithm 1.

Algorithm 1 The Parameter Selection Strategy in Theorem 1, Corollary 1 and 2

- Give initial conditions, topological structure, network parameters
 - Calculate $\varpi, \varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4, \mathcal{H}, \mathcal{H}_1, \mathcal{H}_2$ by LMI toolbox
 - ◊ Ensure $\Xi < 0$ and $\frac{\delta^2}{2\varepsilon_4\pi^2}\varpi - \mathcal{D} \otimes \mathbf{I}_{\mathcal{N}} \leq 0$ in case of strict passivity
 - ◊ Ensure $\tilde{\Xi} < 0$ and $\frac{\delta^2}{2\varepsilon_4\pi^2}\varpi - \mathcal{D} \otimes \mathbf{I}_{\mathcal{N}} \leq 0$ in case of input-strict passivity
 - ◊ Ensure $\hat{\Xi} < 0$ and $\frac{\delta^2}{2\varepsilon_4\pi^2}\varpi - \mathcal{D} \otimes \mathbf{I}_{\mathcal{N}} \leq 0$ in case of output-strict passivity
 - Draw the simulation results
-

3.2 Synchronization for RDNNs

In this part, the asymptotical synchronization of the RDNN (1) is investigated.

Definition 2. The RDNN (1) is asymptotically synchronized if

$$\lim_{t \rightarrow +\infty} \|e_i(t, \cdot)\|_{[\underline{\alpha}, \bar{\alpha}]} = 0, \quad i \in \overline{1, \mathcal{N}}.$$

Theorem 2. Under Assumptions 1-3 and spatial sampled-data controller (7), the RDNN (1) with $\hat{\omega}(t, \varsigma) \equiv \mathbf{0}_{n_{\mathcal{N}}}$ is asymptotically synchronized if the error system (5) is output-strictly passive and $\mathcal{P}^T[\mathcal{H}_2]^s \mathcal{P} > 0$.

Proof. Construct the same Lyapunov functional $V(t)$ as Theorem 1. Since error system (5) is output-strict passive,

$$\dot{V}(t) \leq \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{y}^T(t, \varsigma) \mathcal{H} \hat{\omega}(t, \varsigma) d\varsigma - \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{\omega}^T(t, \varsigma) \mathcal{H}_1 \hat{\omega}(t, \varsigma) d\varsigma - \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{y}^T(t, \varsigma) \mathcal{H}_2 \hat{y}(t, \varsigma) d\varsigma.$$

When $\hat{\omega}(t, \varsigma) = \mathbf{0}_{n_{\mathcal{N}}}$, from the output (6),

$$\dot{V}(t) \leq - \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \mathcal{P}^T[\mathcal{H}_2]^s \mathcal{P} \hat{e}(t, \varsigma) d\varsigma < 0,$$

which ensures that the asymptotical synchronization for the RDNN (1) is realized. $\square$

Theorem 3. Under Assumptions 1-3 and spatial sampled-data controller (7), the RDNN (1) with $\hat{\omega}(t, \varsigma) \equiv \mathbf{0}_{n_{\mathcal{N}}}$ is asymptotically synchronized if there exist scalars $\varepsilon_i > 0$ ($i \in \overline{1, 4}$) such that $\Psi < 0$ and $\frac{\delta^2}{2\varepsilon_4\pi^2}\varpi - \mathcal{D} \otimes \mathbf{I}_{\mathcal{N}} \leq 0$.

Proof. Similar to Theorem 1, from the condition $\Psi < 0$,

$$\dot{V}(t) \leq \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \Psi \hat{e}(t, \varsigma) d\varsigma < 0.$$

which demonstrates that the RDNN (1) is asymptotically synchronized. $\square$

To further reduce the control gain ϖ determined by the condition (C_1) in Theorem 1, there arises a necessity for an adaptive spatial sampled-data control strategy to dynamically adjust the control gain. The adaptive control strategy is designed as

$$\begin{cases} u_i^k(t, \varsigma) = -\varpi_i(t, \varsigma) e_i^k(t, \bar{\varsigma}_p), \\ \frac{\partial \varpi_i(t, \varsigma)}{\partial t} = e^{\beta t} \delta_i \sum_{k=1}^n \xi_i^k e_i^k(t, \varsigma) e_i^k(t, \bar{\varsigma}_p), \end{cases} \quad (15)$$

where $i \in \overline{1, \mathcal{N}}$, $\varpi_i(0, \varsigma) \geq 0$, $\bar{\varsigma}_p = \frac{\varsigma_p + \varsigma_{p+1}}{2}$, $\varsigma \in [\varsigma_p, \varsigma_{p+1})$, $p \in \overline{0, m-1}$, $k \in \overline{1, n}$, δ_i is an arbitrary positive constant and $\beta > 0$ is a sufficiently small real number.

Theorem 4. Under Assumptions 1-3 and the adaptive spatial sampled-data controller (15), the RDNN (1) with $\hat{\omega}(t, \varsigma) \equiv \mathbf{0}_{n\mathcal{N}}$ is asymptotically synchronized.

Proof. Pick a new Lyapunov functional

$$\mathcal{W}(t) = \mathcal{V}(t) + R(t), \quad (16)$$

where

$$\begin{aligned} \mathcal{V}(t) &= e^{\beta t} \mathcal{U}(t) + \eta_1 e^{\beta \bar{\varrho}} \int_{\varrho(t)}^t e^{\beta v} \mathcal{U}(v) dv + \eta_2 e^{\beta \bar{t}} \int_{-\bar{t}}^0 \left[\int_{t+\theta}^t e^{\beta v} \mathcal{U}(v) dv \right] d\theta, \\ R(t) &= \sum_{i=1}^{\mathcal{N}} \frac{1}{2\delta_i} \int_{\underline{\alpha}}^{\bar{\alpha}} \left(\varpi_i(t, \varsigma) - \varpi_i^* \right)^2 d\varsigma, \end{aligned}$$

here $\eta_1 = \frac{\varepsilon_2 L_2}{\bar{\varrho}}$, $\eta_2 = \varepsilon_3 L_3 \bar{t}$, the definition of $\mathcal{U}(t)$ is same as Theorem 1, and $\varpi_i^* > 0$ will be determined in subsequent analysis.

In analogy to Theorem 1, based on the controller (15) and $\hat{\omega}(t, \varsigma) \equiv \mathbf{0}_{n\mathcal{N}}$,

$$\begin{aligned} \dot{\mathcal{U}}(t) &\leq \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \left\{ [\Lambda \mathcal{G}]^s - \Lambda \left[\left(\mathcal{A} - \sum_{r=1}^3 \frac{\mathcal{B}_r \mathcal{B}_r^T}{2\varepsilon_r} - \frac{\varepsilon_1 L_1}{2} \mathbf{I}_n \right) \otimes \mathbf{I}_{\mathcal{N}} \right] \right\} \hat{e}(t, \varsigma) d\varsigma \\ &\quad - \int_{\underline{\alpha}}^{\bar{\alpha}} \left(\frac{\partial \hat{e}(t, \varsigma)}{\partial \varsigma} \right)^T \Lambda (\mathcal{D} \otimes \mathbf{I}_{\mathcal{N}}) \left(\frac{\partial \hat{e}(t, \varsigma)}{\partial \varsigma} \right) d\varsigma \\ &\quad - \sum_{i=1}^{\mathcal{N}} \sum_{k=1}^n \int_{\underline{\alpha}}^{\bar{\alpha}} e_i^k(t, \varsigma) \xi_i^k \varpi_i(t, \varsigma) e_i^k(t, \bar{\varsigma}_p) d\varsigma + \frac{\varepsilon_2 L_2}{2} \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(\varrho(t), \varsigma) \Lambda \hat{e}(\varrho(t), \varsigma) d\varsigma \\ &\quad + \frac{\varepsilon_3 L_3 \bar{t}}{2} \int_{\underline{\alpha}}^{\bar{\alpha}} \int_{t-\bar{t}(t)}^t \hat{e}^T(v, \varsigma) \Lambda \hat{e}(v, \varsigma) dv d\varsigma. \end{aligned} \quad (17)$$

Then, from Assumption 2, one has $\bar{\varrho} + \varrho(t) \geq t$ and

$$\dot{\mathcal{V}}(t) = \beta e^{\beta t} \mathcal{U}(t) + e^{\beta t} \dot{\mathcal{U}}(t) + \eta_1 e^{\beta(\bar{\varrho}+t)} \mathcal{U}(t) - \eta_1 e^{\beta(\bar{\varrho}+\varrho(t))} \mathcal{U}(\varrho(t)) \dot{\varrho}(t)$$

$$\begin{aligned}
& + \eta_2 \bar{t} e^{\beta \bar{t}} e^{\beta t} \mathcal{U}(t) - \eta_2 e^{\beta t} \int_{-\bar{t}}^0 e^{\beta(\bar{t}+\theta)} \mathcal{U}(t+\theta) d\theta \\
& \leq e^{\beta t} \left\{ \dot{\mathcal{U}}(t) + (\beta + \eta_1 e^{\beta \bar{t}} + \eta_2 \bar{t} e^{\beta \bar{t}}) \mathcal{U}(t) - \eta_1 \mathcal{U}(\varrho(t)) \hat{\varrho} - \eta_2 \int_{t-\iota(t)}^t \mathcal{U}(v) dv \right\} \\
& = e^{\beta t} \left\{ \dot{\mathcal{U}}(t) + \frac{\beta + \eta_1 e^{\beta \bar{t}} + \eta_2 \bar{t} e^{\beta \bar{t}}}{2} \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \Lambda \hat{e}(t, \varsigma) d\varsigma \right. \\
& \quad \left. - \frac{\eta_1 \hat{\varrho}}{2} \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(\varrho(t), \varsigma) \Lambda \hat{e}(\varrho(t), \varsigma) d\varsigma - \frac{\eta_2}{2} \int_{\underline{\alpha}}^{\bar{\alpha}} \int_{t-\iota(t)}^t \hat{e}^T(v, \varsigma) \Lambda \hat{e}(v, \varsigma) dv d\varsigma \right\}. \quad (18)
\end{aligned}$$

Thus, in view of the formulas (17) and (18),

$$\begin{aligned}
\dot{\mathcal{W}}(t) & \leq e^{\beta t} \left\{ \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \left\{ [\Lambda \mathcal{G}]^s - \Lambda \left[\left(\mathcal{A} - \sum_{r=1}^3 \frac{\mathcal{B}_r \mathcal{B}_r^T}{2\varepsilon_r} - \left(\frac{\varepsilon_1 L_1 + \beta + L_3 e^{\beta \bar{t}} \bar{t}^2}{2} \right. \right. \right. \right. \right. \\
& \quad \left. \left. \left. + \frac{\varepsilon_2 L_2 e^{\beta \bar{t}}}{2\hat{\varrho}} \right) \mathbf{I}_n \right] \otimes \mathbf{I}_{\mathcal{N}} \right\} \hat{e}(t, \varsigma) d\varsigma - \int_{\underline{\alpha}}^{\bar{\alpha}} \left(\frac{\partial \hat{e}(t, \varsigma)}{\partial \varsigma} \right)^T \Lambda (\mathcal{D} \otimes \mathbf{I}_{\mathcal{N}}) \left(\frac{\partial \hat{e}(t, \varsigma)}{\partial \varsigma} \right) d\varsigma \\
& \quad \left. - \varpi^* \sum_{i=1}^{\mathcal{N}} \sum_{k=1}^n \int_{\underline{\alpha}}^{\bar{\alpha}} \xi_i^k e_i^k(t, \varsigma) e_i^k(t, \bar{\varsigma}_p) d\varsigma \right\}. \quad (19)
\end{aligned}$$

In line with the analysis in the inequality (11), one has

$$\begin{aligned}
& - \varpi^* \sum_{i=1}^{\mathcal{N}} \sum_{k=1}^n \int_{\underline{\alpha}}^{\bar{\alpha}} \xi_i^k e_i^k(t, \varsigma) e_i^k(t, \bar{\varsigma}_p) d\varsigma \\
& \leq - \varpi^* \left(1 - \frac{\varepsilon_4}{2} \right) \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \Lambda \hat{e}(t, \varsigma) d\varsigma + \frac{\delta^2 \varpi^*}{2\varepsilon_4 \pi^2} \int_{\underline{\alpha}}^{\bar{\alpha}} \left(\frac{\partial \hat{e}(t, \varsigma)}{\partial \varsigma} \right)^T \Lambda \frac{\partial \hat{e}(t, \varsigma)}{\partial \varsigma} d\varsigma. \quad (20)
\end{aligned}$$

Substituting inequality (20) into inequality (19) yields that

$$\begin{aligned}
\dot{\mathcal{W}}(t) & \leq e^{\beta t} \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{e}^T(t, \varsigma) \Upsilon_1 \hat{e}(t, \varsigma) d\varsigma + e^{\beta t} \int_{\underline{\alpha}}^{\bar{\alpha}} \left(\frac{\partial \hat{e}(t, \varsigma)}{\partial \varsigma} \right)^T \Upsilon_2 \left(\frac{\partial \hat{e}(t, \varsigma)}{\partial \varsigma} \right) d\varsigma \\
& = e^{\beta t} \int_{\underline{\alpha}}^{\bar{\alpha}} \tilde{\mathbf{E}}^T(t, \varsigma) \Theta \tilde{\mathbf{E}}(t, \varsigma) d\varsigma,
\end{aligned}$$

where $\tilde{\mathbf{E}}(t, \varsigma) = \left(\hat{e}(t, \varsigma), \frac{\partial \hat{e}(t, \varsigma)}{\partial t} \right)^T$, $\Upsilon_1 = [\Lambda \mathcal{G}]^s - \Lambda \left[\left(\mathcal{A} - \sum_{r=1}^3 \frac{\mathcal{B}_r \mathcal{B}_r^T}{2\varepsilon_r} - \left[\frac{\varepsilon_1 L_1 + \beta + \varepsilon_3 L_3 e^{\beta \bar{t}} \bar{t}^2}{2} + \frac{\varepsilon_2 L_2 e^{\beta \bar{t}}}{2\hat{\varrho}} - \left(1 - \frac{\varepsilon_4}{2} \right) \varpi^* \right] \mathbf{I}_n \right) \otimes \mathbf{I}_{\mathcal{N}} \right]$, $\Upsilon_2 = \Lambda \left[\left(\frac{\delta^2 \varpi^*}{2\varepsilon_4 \pi^2} \mathbf{I}_n - \mathcal{D} \right) \otimes \mathbf{I}_{\mathcal{N}} \right]$ and

$$\Theta = \begin{pmatrix} \Upsilon_1 & 0 \\ 0 & \Upsilon_2 \end{pmatrix}.$$

By selecting ϖ^* , ε_1 , ε_2 , ε_3 and ε_4 appropriately such that $\Theta \leq 0$, one has $\dot{\mathcal{W}}(t) \leq 0$, which yields $\mathcal{W}(t) \leq \mathcal{W}(0)$.

On the other hand,

$$\mathcal{W}(t) \geq \mathcal{V}(t) \geq e^{\beta t} \frac{\lambda_{\min}(A)}{2} \|e(t, \cdot)\|_{[\underline{\alpha}, \bar{\alpha}]}^2. \quad (21)$$

So,

$$\|e(t, \cdot)\|_{[\underline{\alpha}, \bar{\alpha}]} \leq \kappa \sqrt{\mathcal{W}(0)} e^{-\frac{\beta}{2}t}, \quad t \geq 0,$$

where $\kappa = \sqrt{\frac{2}{\lambda_{\min}(A)}}$. It implies that the RDNN (1) is asymptotically synchronized. $\square$

Remark 4. In [39], Barbalat's lemma was used to prove the asymptotic synchronization of CNNs without diffusion terms under adaptive control. However, this analytical method cannot be directly applied to the synchronization analysis of RDNNs with mixed delays, because the existence of the diffusion term makes it impossible to guarantee the uniform continuity of $\mathcal{W}(t)$. Additionally, in [16], based on reduction to absurdity, the asymptotic synchronization of RDNNs without delay was ensured under an adaptive strategy. Unfortunately, this method is also inapplicable to the RDNNs considered in this article due to the existence of mixed delays. Inspired by [29, 30], an exponential term $e^{\beta t}$ was introduced into the construction of LKF and control design, the asymptotic synchronization of delayed RDNNs under adaptive control was successfully solved in Theorem 4.

4 Numerical Simulations

Two examples are provided to verify the passivity and synchronization results in this section.

Table 2: Parameter selection

Delays and Nonlinear Activation Functions
$\varrho(t) = t - \frac{\exp(t)}{1+\exp(t)},$ with $\hat{\varrho} = 0.75,$ $\iota(t) = 0.5 + 0.2 \sin(2\pi t),$ with $\bar{\iota} = 0.7,$ $f_1(z_i) = f_3(z_i) = (\tanh(z_i^1), \tanh(z_i^2), \tanh(z_i^3))^T,$ $f_2(z_i) = 2(\tanh(z_i^1), \tanh(z_i^2), \tanh(z_i^3))^T.$
The Initial and Boundary Conditions
$z(t, \varsigma)$ are given as arbitrary constants in $[-60, 60], (t, \varsigma) \in [-0.75, 0] \times (-5, 5),$ $z_i(t, -5) = z_i(t, 5) = \mathbf{0}_3.$
The Input and Output Vectors
$y_i(t, \varsigma) = P e_i(t, \varsigma) + Q \bar{\omega}_i(t, \varsigma), i \in \overline{1, 5}$ with $P = 4\mathbf{I}_3$ and $Q = 5\mathbf{I}_3.$ $\omega_i(t, \varsigma) = \begin{pmatrix} 0.07i \sin(\frac{\pi}{2} + x) \cos(0.01t) \\ 0.09i \sin(\frac{\pi}{2} + x) \cos(0.005t) \\ 0.05i \sin(\frac{\pi}{2} + x) \cos(0.003t) \end{pmatrix}.$

Example 1. Consider a kind of RDNNs with mixed time-varying delays and 3-weighted couplings, whose dynamics satisfies equation (1), where $t \in \mathcal{R}_{\geq 0}$, $\varsigma \in [-5, 5]$, $\mathcal{D} = \mathcal{A} = 0.6\mathbf{I}_3$, $c^1 = c^3 = 0.1$, $c^2 = 0.2$, $\Gamma^1 = \text{diag}\{0.1, 0.2, 0.1\}$, $\Gamma^2 = \text{diag}\{0.3, 0.2, 0.3\}$, $\Gamma^3 = \text{diag}\{0.2, 0.1, 0.2\}$, the corresponding functions are given by Table 2, $\varsigma \in [-5 + 0.1p, -5 + 0.1(p + 1)]$ ($p \in \overline{0, 99}$) are the spatial sampling intervals, and the coefficient matrices

$$B_1 = \begin{bmatrix} 1 & 5 & 7 \\ -0.5 & 1 & 1.2 \\ 9 & 4 & -0.8 \end{bmatrix}, \quad B_2 = \begin{bmatrix} -0.1 & 3 & 5 \\ -2 & -0.2 & 1 \\ -2 & -5 & -0.1 \end{bmatrix}, \quad B_3 = \begin{bmatrix} 0.1 & -3 & -4 \\ 0.2 & 0.3 & -6 \\ 5 & 4 & 0.2 \end{bmatrix}.$$

The topology is given in Fig. 1, the weighted union communication topology is evidently strong connected with $\xi^{(1)} = \xi^{(2)} = \xi^{(3)} = 0.2\mathbf{1}_5$. By simple calculation, $L_1 = L_3 = 1$, $L_2 = 2$.

To verify the passivity of the error system (5), define

$$\begin{aligned} \psi_1(t) &= \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{y}^T(t, \varsigma) \mathcal{H} \hat{\omega}(t, \varsigma) d\varsigma - \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{\omega}^T(t, \varsigma) \mathcal{H}_1 \hat{\omega}(t, \varsigma) d\varsigma - \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{y}^T(t, \varsigma) \mathcal{H}_2 \hat{y}(t, \varsigma) d\varsigma - \dot{V}(t), \\ \psi_2(t) &= \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{y}^T(t, \varsigma) \mathcal{H} \hat{\omega}(t, \varsigma) d\varsigma - \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{\omega}^T(t, \varsigma) \mathcal{H}_1 \hat{\omega}(t, \varsigma) d\varsigma - \dot{V}(t), \\ \psi_3(t) &= \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{y}^T(t, \varsigma) \mathcal{H} \hat{\omega}(t, \varsigma) d\varsigma - \int_{\underline{\alpha}}^{\bar{\alpha}} \hat{y}^T(t, \varsigma) \mathcal{H}_2 \hat{y}(t, \varsigma) d\varsigma - \dot{V}(t). \end{aligned}$$

Case I. To satisfy the conditions C_1 and C_2 in Theorem 1, choosing $\varepsilon_1 = \varepsilon_2 = \varepsilon_3 = 7$, $\varepsilon_4 = 2$, and by utilizing MATLAB LMI toolbox,

$$\begin{aligned} \mathcal{H} &= \begin{bmatrix} 12.5585 & 0.1938 & -0.1117 \\ 0.1938 & 12.1270 & 0.0632 \\ -0.1117 & 0.0632 & 12.7263 \end{bmatrix} \otimes \mathbf{I}_5, \quad \mathcal{H}_1 = \begin{bmatrix} 17.6768 & 0.2102 & -0.1839 \\ 0.2353 & 17.1807 & 0.0642 \\ -0.0729 & 0.0813 & 17.8698 \end{bmatrix} \otimes \mathbf{I}_5, \\ \mathcal{H}_2 &= \begin{bmatrix} 1.0974 & -19.8240 & 95.3621 \\ 19.8611 & 1.0562 & 10.5091 \\ 95.3833 & -10.4970 & 1.1134 \end{bmatrix} \otimes \mathbf{I}_5, \end{aligned}$$

and the control gain $\varpi = 12.2429\mathbf{I}_5$. Under the spatial sampled-data controller (7), from Theorem 1, the strict passivity of error system (5) is achieved and shown in Fig. 3.

Case II. When $\mathcal{H}_2 = \mathbf{0}_{15}$, choose $\varepsilon_1 = \varepsilon_3 = 9$, $\varepsilon_2 = 6$, $\varepsilon_4 = 0.8$. By employing the MATLAB LMI toolbox, $\mathcal{H}, \mathcal{H}_1$ are solved as

$$\mathcal{H} = \begin{bmatrix} 0.1648 & -0.0692 & 0.0493 \\ -0.0692 & 0.3336 & -0.0254 \\ 0.0493 & -0.0254 & 0.1136 \end{bmatrix} \otimes \mathbf{I}_5, \quad \mathcal{H}_1 = \begin{bmatrix} 0.3278 & 31.5829 & 37.4941 \\ -31.8403 & 0.6412 & -20.3522 \\ -37.3136 & 20.2572 & 0.2337 \end{bmatrix} \otimes \mathbf{I}_5,$$

and control gain $\varpi = 36.1213\mathbf{I}_5$. Under the spatial sampled-data controller (7), the error system (5) is input-strictly passive and shown in Fig. 4.

Case III. When $\mathcal{H}_1 = \mathbf{0}_{15}$, select $\varepsilon_1 = 3$, $\varepsilon_2 = 6$, $\varepsilon_3 = 2$, and $\varepsilon_4 = 1.7$. By employing the MATLAB LMI toolbox, $\mathcal{H}$, $\mathcal{H}_2$ are calculated as

$$\mathcal{H} = \begin{bmatrix} 8.0862 & 0.9488 & -0.4061 \\ 0.9488 & 6.7500 & 0.2155 \\ -0.4061 & 0.2155 & 8.9455 \end{bmatrix} \otimes \mathbf{I}_5, \mathcal{H}_2 = \begin{bmatrix} 0.7562 & 898.0732 & 455.6378 \\ -897.8896 & 0.6270 & -4.8870 \\ -455.7162 & 4.9288 & 0.8392 \end{bmatrix} \otimes \mathbf{I}_5.$$

and control gain $\varpi = 11.4197\mathbf{I}_5$. Based on Corollary 2, under the spatial sampled-data controller (7), the error system (5) is output-strictly passive and shown in Fig. 5.

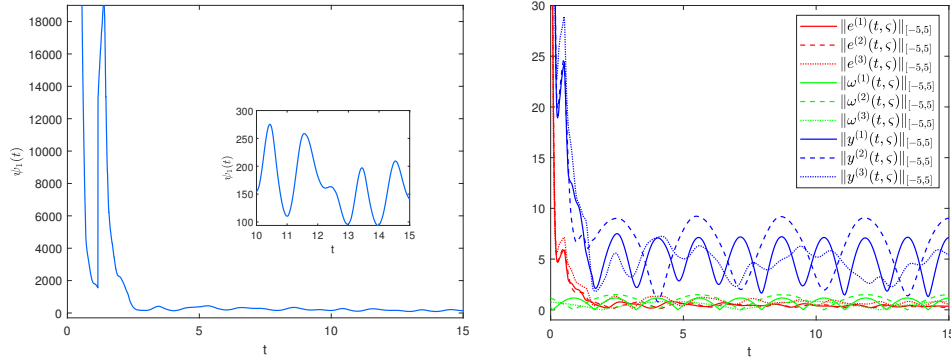


Fig. 3: Evolution of function $\psi_1(t)$ and the error $e_i^k(t, \varsigma)$ in Case I.

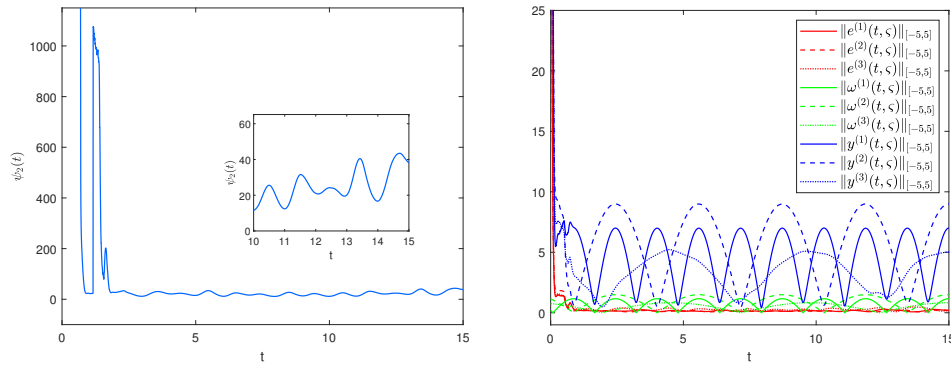


Fig. 4: Evolution of function $\psi_2(t)$ and the error $e_i^k(t, \varsigma)$ in Case II.

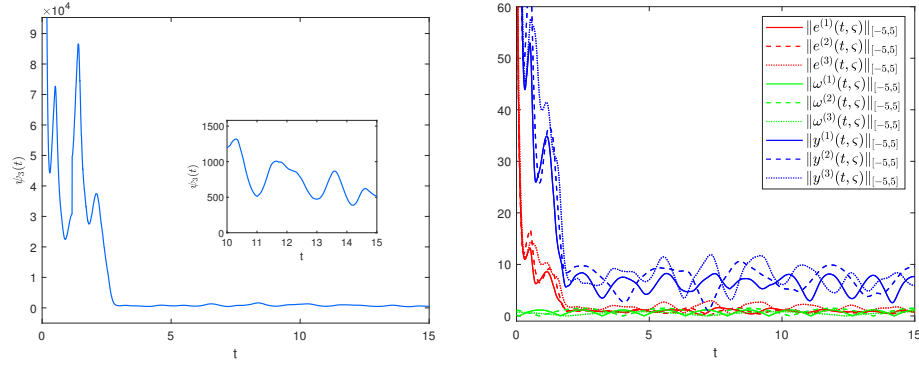


Fig. 5: Evolution of function $\psi_3(t)$ and the error $e_i^k(t, \varsigma)$ in Case III.

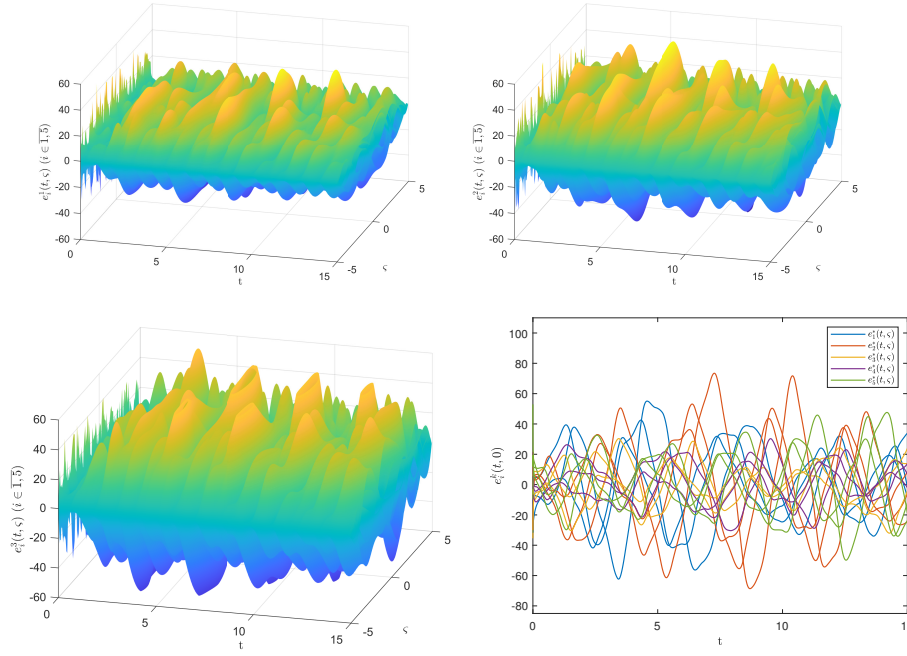


Fig. 6: Spatiotemporal evolution of $e_i^k(t, \varsigma)$ without control.

Next, the asymptotic synchronization of the RDNN (1) is verified. Fig. 6 shows that the RDNN (1) fails to be synchronized in the absence of control. Under the spatial sampled-data controller (7), the error system (5) is output-strictly passive in Case III. When $\hat{\omega}(t, \varsigma) \equiv \mathbf{0}_{15}$, the network (1) is synchronized, as ensured by Theorem 2 and illustrated by Fig. 7.

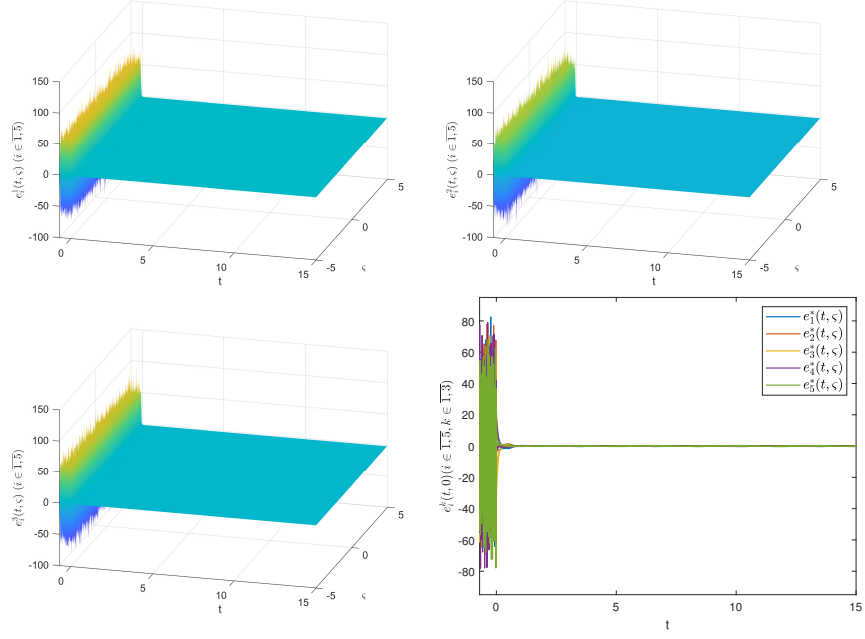


Fig. 7: Synchronization evolution under the spatial sampled-data controller (7).

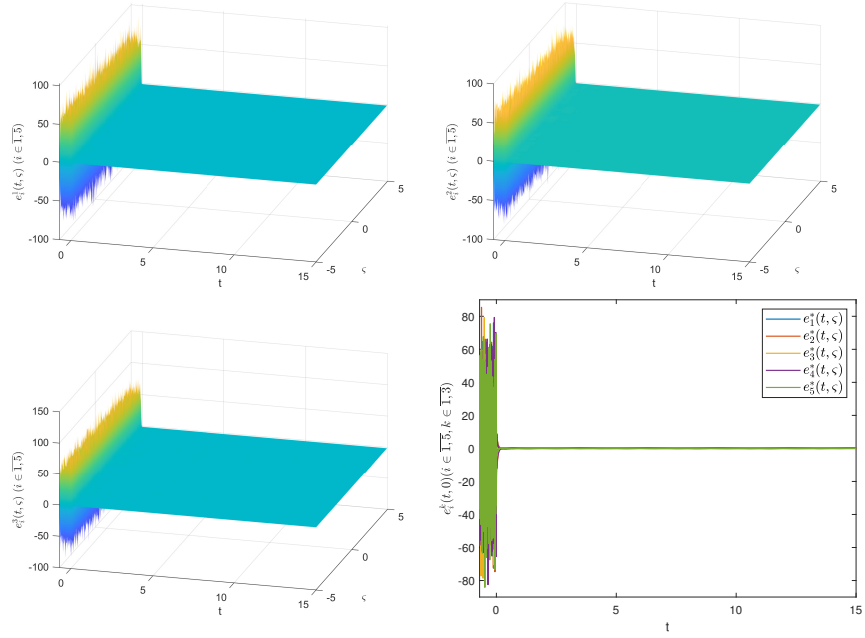


Fig. 8: Synchronization evolution under controller (15).

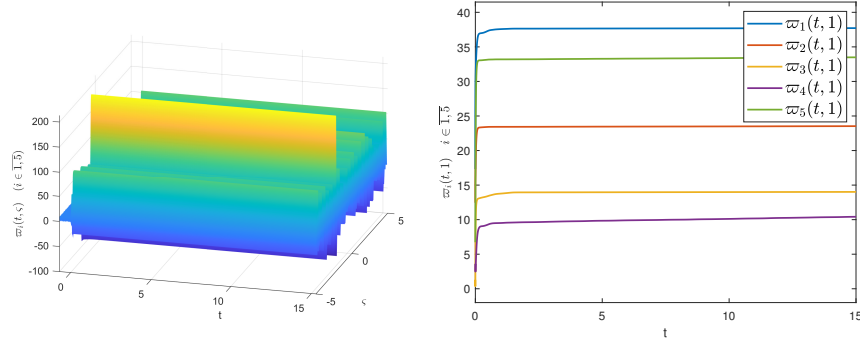


Fig. 9: Control gains $\varpi_i(t, \varsigma)$ in controller (15), $i \in \overline{1, 5}$.

In what follows, the asymptotic synchronization of system (1) under adaptive spatial sampled-data controller (15) is verified. Choose $\beta = 10^{-3}$, $\delta_i = 10$ and the initial conditions $\varpi_i(t, \varsigma)$ are given as the arbitrary constants in $[0, 10]$ ($i \in \overline{1, 5}$, $(t, \varsigma) \in [-0.75, 0] \times (-5, 5)$). The asymptotic synchronization of the RDNN (1) with $\hat{\omega}(t, \varsigma) \equiv \mathbf{0}_{15}$ is achieved and shown in Fig. 8, and the dynamic of control gain $\varpi_i(t, \varsigma)$ is illustrated by Fig. 9.

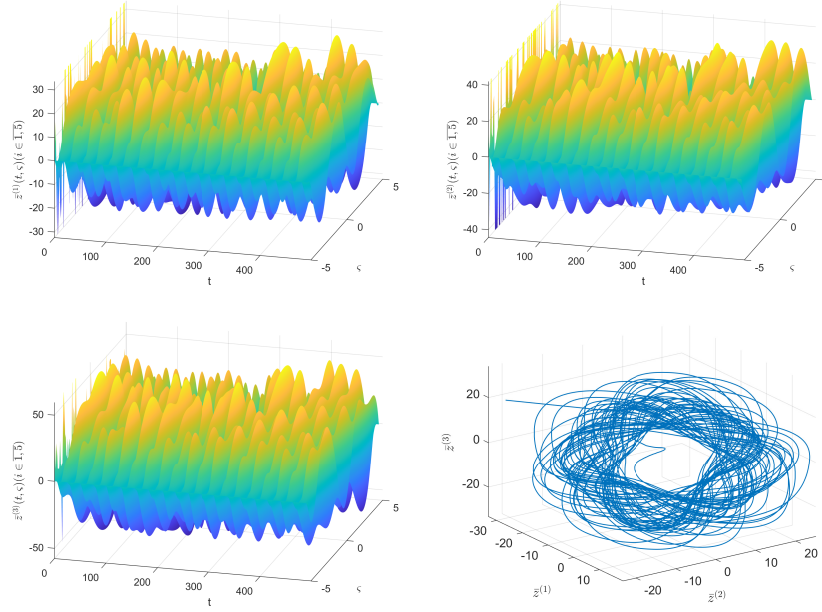


Fig. 10: The spatiotemporal chaotic behavior of the RDNN (1).

Example 2. Based on the chaotic Logistic encryption scheme with the scrambling operation in [43], the passivity-based synchronization of RDNN (1) and the corresponding chaotic sequence are applied to image encryption and decryption via spatial sampled-data

controller (7). The spatiotemporal chaotic behavior of $\bar{z}$ in the RDNN (1) is shown in Fig. 10.

The RDNN (1) and the weighted-average state (4) are regarded as receivers and sender, respectively. According to Example 1, the RDNN (1) is synchronized after $T_0 = 5$. The detailed steps of the encryption algorithm are provided as follows.

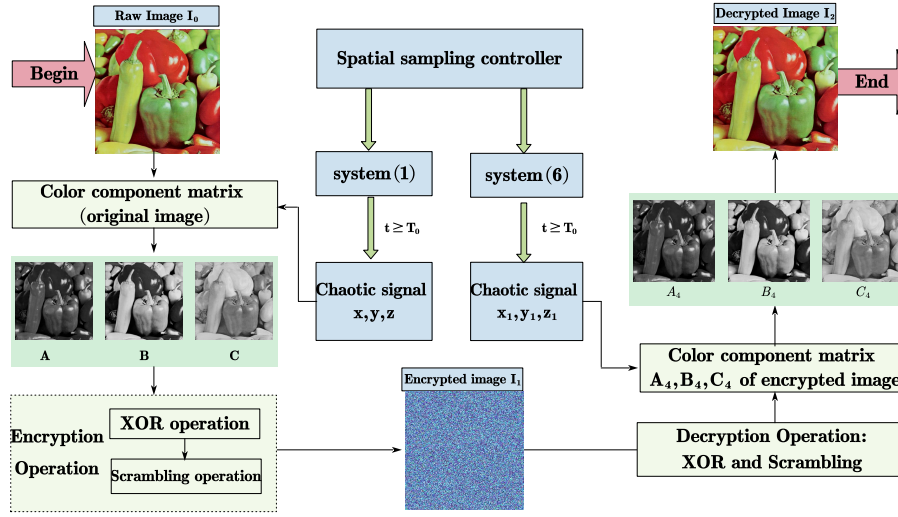


Fig. 11: Flow of image encryption and decryption.

Step 1. Original image process. Read the original image I_0 and resize it to 256×256 . Separate color image I_0 into red, green, and blue components so that three pixel series are formulated, $A(h, j)$, $B(h, j)$, $C(h, j)$, $h \in \overline{1, 256}$, $j \in \overline{1, 256}$.

Step 2. Chaotic signal process. The chaotic sequences generated in system (4) are transformed to three security channels and denoted as x, y, z . Then, amplify the chaotic signals at a certain scale,

$$X = 10000(x + 5), Y = 10000(y + 5), Z = 10000(z + 5).$$

To facilitate the XOR operation, the scaled chaotic signals X, Y, Z are converted into 8-bit integers X_0, Y_0, Z_0 by using the *im2uint8* function

$$X_0 = \text{im2uint8}(X), Y_0 = \text{im2uint8}(Y), Z_0 = \text{im2uint8}(Z).$$

Subsequently, rearranging the chaotic sequences X, Y, Z to generate three new indexed sequences l_x, l_y, l_z . The color matrices A, B, C of the original image are reshaped into three vectors A_1, B_1, C_1 , i.e.,

$$A_1 = \text{reshape}(A, 1, n\mathcal{N}), B_1 = \text{reshape}(B, 1, n\mathcal{N}), C_1 = \text{reshape}(C, 1, n\mathcal{N}).$$

Step 3. Encryption. The encrypted pixel matrices of original image are generated by applying the following XOR operation

$$A_2 = A_1 \oplus X_0, \quad B_2 = B_1 \oplus Y_0, \quad C_2 = C_1 \oplus Z_0.$$

Then, rearranging the pixel values in A_2, B_2, C_2 based on indices l_x, l_y, l_z . Finally, the encrypted image I_1 is obtained by

$$A_3 = \text{reshape}(A_2, n, \mathcal{N}), \quad B_3 = \text{reshape}(B_2, n, \mathcal{N}), \quad C_3 = \text{reshape}(C_2, n, \mathcal{N}).$$

The decryption procedures are outlined as follows.

Step 1. Original image process. Read the original image I_1 . Separate color image I_1 into red, green, and blue components so that three pixel series are yielded, $A_4(h, j)$, $B_4(h, j)$, $C_4(h, j)$, $h \in \overline{1, 256}$, $j \in \overline{1, 256}$.

Step 2. Chaotic signal process. The chaotic sequences generated in system (1) are transformed to three security channels and denoted as x_1, y_1, z_1 . Then, amplify the chaotic signals at a certain scale,

$$X_1 = 10000(x_1 + 5), \quad Y_1 = 10000(y_1 + 5), \quad Z_1 = 10000(z_1 + 5).$$

To facilitate the XOR operation, the scaled chaotic signals X_1, Y_1, Z_1 are converted into 8-bit integers $X_0^{(1)}, Y_0^{(1)}, Z_0^{(1)}$ by using the *im2uint8* function

$$X_0^{(1)} = \text{im2uint8}(X_1), \quad Y_0^{(1)} = \text{im2uint8}(Y_1), \quad Z_0^{(1)} = \text{im2uint8}(Z_1).$$

Subsequently, rearranging the chaotic sequences X_1, Y_1, Z_1 to generate three new indexed sequences $l_x^{(1)}, l_y^{(1)}, l_z^{(1)}$. The color matrices A_4, B_4, C_4 of the original image are reshaped into three vectors

$$A_5 = \text{reshape}(A_4, 1, n\mathcal{N}), \quad B_5 = \text{reshape}(B_4, 1, n\mathcal{N}), \quad C_5 = \text{reshape}(C_4, 1, n\mathcal{N}).$$

Step 3. Decryption. The encrypted pixel matrices of original image are generated by applying the following XOR operation

$$A_6 = A_5 \oplus X_0^{(1)}, \quad B_6 = B_5 \oplus Y_0^{(1)}, \quad C_6 = C_5 \oplus Z_0^{(1)}.$$

Then, rearrange the pixel values in A_6, B_6, C_6 utilizing indices $l_x^{(1)}, l_y^{(1)}, l_z^{(1)}$. Finally, the decrypted image I_3 is obtained by

$$A_7 = \text{reshape}(A_6, n, \mathcal{N}), \quad B_7 = \text{reshape}(B_6, n, \mathcal{N}), \quad C_7 = \text{reshape}(C_6, n, \mathcal{N}).$$

A 512×512 grayscale pepper image is selected as the test image shown in Fig.12. The encrypted image is illustrated in Fig.13, where the underlying image becomes indistinguishable. Through a series of reverse operations, the image is successfully decrypted, as depicted in Fig.14. Obviously, the decrypted image Fig.14 is identical to the original image Fig. 12, which implies that the proposed passivity-based synchronization criteria and controller has reliable performance in image encryption and decryption.



Fig. 12: Original Image.

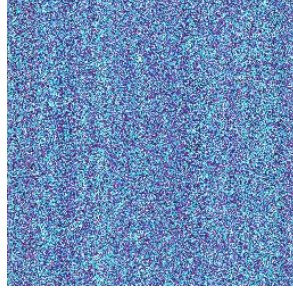


Fig. 13: Encrypted Image.



Fig. 14: Decrypted Image.

5 Conclusion

This article investigated the passivity and synchronization of RDNNs with mixed delays and multi-weighted couplings based on spatial sampling control strategy. By constructing the LKF, several passivity and synchronization criteria were derived by the technique of rearranging variable order and inequality skills. These conditions are determined by the diffusion coefficients, coupled strengths, time delays, sampling interval length and control parameters. An adaptive spatial sampling control scheme was proposed and the adaptive synchronization was rigorously analyzed. This control strategy integrates spatial sampling control with adaptive control, effectively reducing control costs and facilitating practical operation. Furthermore, the proposed passivity-based synchronization criteria and controllers have been applied to image encryption and decryption. Future research will explore the time-limited passivity-based synchronization of RDNNs via spatial sampling control.

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